The Geometry of Generation in the UNNS Substrate: A Mathematical Synthesis

UNNS Research Collective

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Abstract

Two arithmetic transformations of the positive integers and rationals generate geometry in different ways. The first builds integers by composing the maps $m \mapsto m(N+1)^2/N$ with $N \mid m$; the second iterates $x \mapsto x + L^2/G$ on positive rationals, with L and G determined by the integers $d \geq 2$ for which d^2 divides the numerator of x and $d \equiv 1$ modulo its denominator. To each state of either kind one attaches the set D of such integers, an arithmetic lattice of relations, a set of primitive relations and a cell complex, and to each generated integer one attaches its complete histories and the complexes they form. We ask which of these structures determine which others.

We prove four kinds of statement. Projections factor: endpoint, content, swap class and occurrence set form a chain of maps, and every cell-complex invariant of an arithmetic state factors through its occurrence set. Projections lose information, with exact witnesses: equal endpoints with different label content, equal occurrence sets with different primitive relation systems, and equal occurrence sets with different numbers of histories and of swap classes. Evolution is organised by a hierarchy of persistence: occurrence sets grow, the connectivity relation of the realised complex only coarsens, an edge exists during an interval of times determined by two arrival times, and the first two Betti numbers are not monotone along the examined evolutions; the change of the first two Betti numbers at a graph-level event is determined by local counts together with prime supports. Finally, the two transformations are compared at five levels: no one-step identity correspondence between them exists (a general intertwining map remains open), a translation formula transports predecessors along the collective map for integer states, and the remaining constructions are stated as open problems with their nearest proved theorems. Read together, the results support an operational characterisation of the generative architecture: it is described by admissible transformations among states, histories, occurrence inventories and realised complexes, not by any one of these alone; an intrinsic axiomatisation is not claimed. No universal law is claimed.

Contents

1 Introduction and the mathematical synthesis problem **3**

1.1 The problem 3

1.2 The four governing questions 4

1.3 Contributions and what is not claimed 4

1.4 Reading guide 5

2 States, histories, occurrence geometry and evolution **5**
2.1 Generative states and histories . . . 5
2.2 Collective states . . . 6
2.3 Occurrence geometry . . . 6
2.4 Lattices and primitive relations . . . 6
2.5 Factorization . . . 6

3 Numerical projection and generative information **7**
3.1 Cited ingredients . . . 8
3.2 The proposition . . . 8

4 Occurrence projection and dormant arithmetic structure **9**

5 Geometric realization and dimensional transport **11**
5.1 Three tiers of relation . . . 11
5.2 An abstract minimality principle . . . 12
5.3 The dimensional boundary of transport . . . 13

6 Arithmetic constraints and history topology **13**
6.1 Local guards and global classes . . . 13
6.2 Cubes and homology . . . 15

7 Endpoint geometry versus global ancestry **17**

8 Structural changes during collective evolution **18**
8.1 Connectivity is governed by prime supports . . . 18

9 Arithmetic prediction and homological consequences **21**
9.1 What arithmetic predicts . . . 21
9.2 Information requirements . . . 22

10 Collective recurrence and geometric behavior **24**

11 Correspondence between distinct generating rules **24**

12 The unified diagram **26**
12.1 Objects, maps and statuses . . . 26
12.2 Factorizations . . . 26
12.3 Commuting relations . . . 27

13 Synthesis propositions and obstructions **27**
13.1 Answers to S1–S4 . . . 28

14 Relation to established theories **30**

15 Principal mathematical conclusions **31**
15.1 Operational characterisation of the UNNS Substrate 31
15.2 Mathematical implications 32
15.3 Limitations 33

16 Open questions **33**
16.1 Operational versus intrinsic characterisation 34

References **36**

A Notation **37**

B Theorem provenance **37**

C Proofs of the synthesis lemmas **39**

D Diagram definitions **40**

E Counterexamples and obstructions **40**

F Finite-exact information **41**

1 Introduction and the mathematical synthesis problem

1.1 The problem

Fix the two maps

$$T_N(m) = m\frac{(N+1)^2}{N} \quad (N \mid m,\ m, N \in \mathbb{Z}_{\geq 1}), \qquad C(x) = x + \frac{L(x)^2}{G(x)} \quad (x \in \mathbb{Q}_{>0}),$$

where, for $x = a/b$ in lowest terms, $D(x) = \{d \geq 2 :\ d^2 \mid a,\ d \equiv 1 \pmod b\}$, $L = \operatorname{lcm} D(x)$ and $G = \gcd\{d - 1 : d \in D(x)\}$. The first map composes into *histories* $(m_0; N_1, \ldots, N_r)$ that reach an integer; the second is a deterministic dynamics. Both maps are governed by divisibility of a numerator by squares, and both attach to their states the same kind of object: an occurrence set D, an arithmetic lattice of relations among the occurrences, a set of primitive relations in it, and cell complexes built from them. The three technical manuscripts of this programme studied these objects separately: lattice transport and its dimensional obstructions [1], the topology of the spaces of histories [2], and the evolution of the complexes along collective orbits [3]. The present manuscript asks a single question.

> *Which of these structures determine which others, and where exactly do the determinations fail?*

An answer has a definite logical form. A determination is a *factorization*: a map f factors through a map π exactly when f is constant on the fibres of π (Lemma 2.5). A failure of determination is an *obstruction*: two points of the domain with the same image under π and different images under f. Every claim below is of one of these two forms, or is an explicitly marked open construction, or is an interpretive comparison without a claimed correspondence.

What the synthesis says about the Substrate. The mathematical question has a structural counterpart: what do these results say about the generative architecture of the UNNS Substrate itself? The answer given here is deliberately operational. States, histories, occurrence inventories, primitive relations and realised complexes are linked by the proved factorizations, admissibility conditions and obstructions of Sections 12 and 13, and Section 15.1 states what these relations support and what they do not. It is a synthesis conclusion, not a numbered theorem.

1.2 The four governing questions

We retain the four organising questions of the programme and answer each by a proposition, an obstruction, a conditional result or a precise open problem.

S1 (common guards). The admissibility condition $N \mid m$ of a history step and the occurrence condition $d^2 \mid a$, $d \equiv 1\ (b)$ of a collective event both involve divisibility of the integer by squares and congruences. For integer states the two guards coincide exactly (Proposition 4.1(a), a bijection between occurrences and incoming steps); for rational states the congruence $d \equiv 1\ (b)$ has no counterpart in the generative rule, and a rational extension is an open construction (Section 16, O8).

S2 (information lost by projection). The chain endpoint $\leftarrow$ content $\leftarrow$ swap class $\leftarrow$ history, and the chain $x \mapsto D(x) \mapsto$ realised complex, are chains of factorizations; every step loses information, with exact witnesses (Propositions 4.1, 7.1 and 3.4). The data that *do* reconstruct a history are the occurrence sets along the whole path (Theorem 3.2), not the occurrence set of the endpoint.

S3 (geometry across spaces). The complex of an integer state is determined by D; the history space of the same integer has independent arithmetic. The endpoint occurrence complex does not determine the associated history complex or its recombination invariants (Proposition 7.1); a general structure-preserving correspondence remains unestablished. Relations between history spaces and evolving complexes beyond the level of occurrence labels are open (Proposition 11.3, O3).

S4 (boundaries and events). The transport criterion of Paper I and the edge-loss criterion of Paper III are instances of one abstract monotone-minimality principle (Lemma 5.2 and Proposition 5.3); a direct comparison of the two ambient orders is an open problem (O7).

1.3 Contributions and what is not claimed

The new mathematics of this manuscript consists of nine propositions SIV-P1–SIV-P9 with independent proofs (Section 13 gives the register), five lemmas used in them (Appendix C), the global diagram (Figure 21), and exact witnesses recomputed from scratch (Appendix F). Results of Papers I–III are cited with their theorem numbers and crosswalk identifiers (Appendix B); they are not proved again unless a new derivation needs them in a modified form. Computations quoted from those manuscripts keep their original status; the verification of the frozen releases is inherited and was not re-run.

We do *not* claim a universal organising law, a general commuting diagram between the generative and collective sides (Section 12 shows that none of the natural squares is claimed to

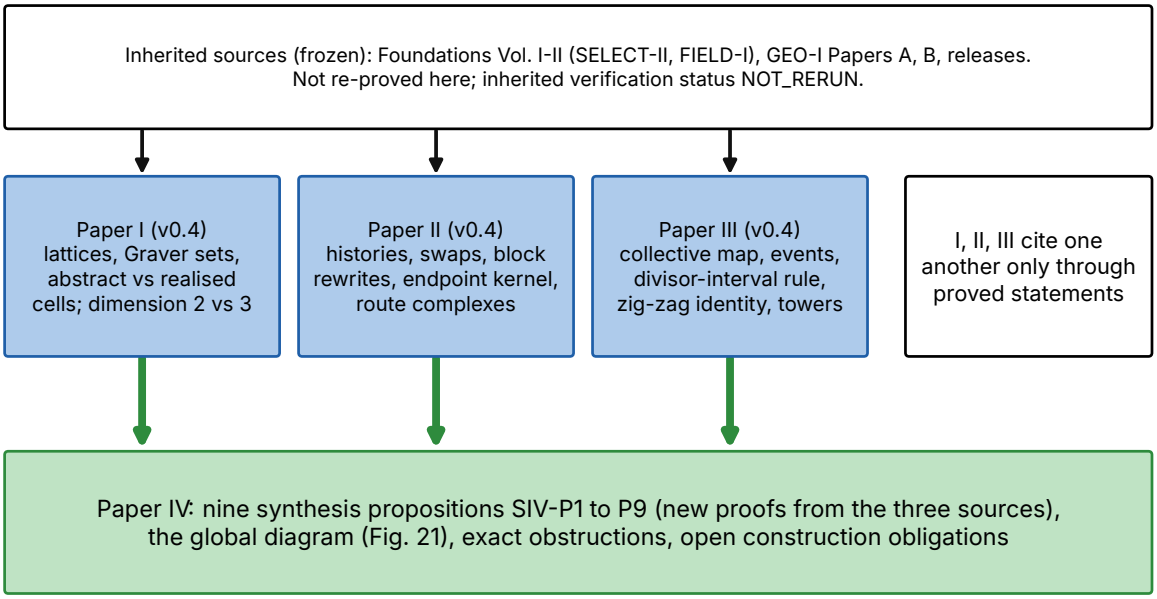


Figure 1: **Architecture of the programme.** Paper IV draws only on proved statements of Papers I–III (v0.4) and on the frozen foundations. *Schematic of inheritance; no mathematical object is drawn.*

commute), or any result about the third constant-type conjecture of the programme (statement C3 remains unclaimed). Finite-exact statements are never extrapolated.

1.4 Reading guide

Section 2 fixes definitions. Sections 3 and 4 treat the two projections; Sections 5 to 7 treat geometry, history topology and their separation; Sections 8 to 10 treat evolution; Section 11 compares the two dynamics; Section 12 presents the global diagram; Sections 13 to 16 collect results, relations to known theory, conclusions and open problems. Arrow conventions in all diagrams: a black solid arrow is an *established arrow*; a thick green arrow is a *proved factorization*; a red arrow with a cross and a bar is a *proved obstruction*; a grey dashed arrow is an *open construction*; a purple dotted double-headed arrow is an *interpretive comparison.* Figure 1 shows architecture of the programme.

2 States, histories, occurrence geometry and evolution

This section fixes the objects, with the exact source of each definition. Notation is collected in Appendix A.

2.1 Generative states and histories

Definition 2.1 (source: Foundations Vol. I; Paper II, Definition 2.1, Paper II, Definition 2.4)**.** For integers $m, N \geq 1$ with $N \mid m$ write $m \xrightarrow{N} T_N(m) = m(N+1)^2/N$. A *history* from

the source m_0 is a word $w = (N_1, \dots, N_r)$ with $N_i \mid m_{i-1}$, $m_i = T_{N_i}(m_{i-1})$; its *endpoint* is $\mathrm{end}(h) = m_r$, its *content* is the multiset $c(h) \in \mathbb{N}^A$ of its letters. An adjacent transposition is *legal* if both words are admissible; the *swap graph* on the admissible arrangements of a fixed content has legal transpositions as edges, and its components are the *swap classes*. The *fibre sequence* of h is $F(h) = (D(m_0), \dots, D(m_r))$.

Since $m_i = m_{i-1}R(N_i)$ with $R(N) = (N+1)^2/N$, the endpoint is $\mathrm{end}(h) = m_0 \prod_N R(N)^{c_N}$ and depends only on (m_0, c).

2.2 Collective states

Definition 2.2 (source: Foundations Vol. II; Paper III, Definition 2.1)**.** For $x = a/b > 0$ in lowest terms, $D(x) = \{d \geq 2 : d^2 \mid a,\ d \equiv 1 \pmod{b}\}$. If $D(x) \neq \emptyset$ put $L = \operatorname{lcm} D(x)$, $G = \gcd\{d-1\}$ and $C(x) = x + L^2/G$. For an integer t the congruence is vacuous and $D(t) = \{d \geq 2 : d^2 \mid t\}$.

Lemma 2.3 (occurrence monotonicity; Paper III, Lemma 2.6)**.** $D(x) \subseteq D(C(x))$. *Hence along every orbit $x_n = C^n(x)$ the sets $D_n = D(x_n)$ are non-decreasing.*

2.3 Occurrence geometry

Definition 2.4 (the divisor-interval complex; Paper III, Definition 9.2, Paper III, Theorem 3.4, Paper III, Theorem 9.1)**.** For a finite set $D \subset \mathbb{Z}_{\geq 2}$ and $u, v \in D$ put $I(u,v) = \{z \in \mathbb{Z} : \gcd(u,v) \mid z \mid \operatorname{lcm}(u,v)\}$. The complex $K(D)$ has vertex set D, an edge $\{u,v\}$ iff $I(u,v) \cap (D \cup \{1\}) = \{u,v\}$, and higher cells (cubical cells spanned by parallel edges) as in Paper III. For a state x, $K_{\mathrm{move}}(x) = K(D(x))$.

By Paper III, Theorem 3.4, the realised fibre, $K_{\mathrm{move}}(x)$, its boundary maps, components and homology are functions of $D(x)$ alone. For an integer t, the active primes are those with $p^2 \mid t$ and $K_{\mathrm{move}}(t)$ is contractible [3, Paper III, Theorem 3.1].

2.4 Lattices and primitive relations

For a state $x = a/b$ with active primes $p_1, \dots, p_k$ (those p with $p^2 \mid a$ and $p \nmid b$) let $\Lambda_b \subseteq \mathbb{Z}^k$ be the lattice of exponent vectors α with $\prod p_i^{\alpha_i} \equiv 1 \pmod{b}$, let F be the fibre (its points in the box of admissible exponents, corresponding to $D(x)$) and $\mathrm{Gr}_q(x)$ the region Graver set: the primitive (conformally minimal) lattice vectors in the box. These are the objects of Paper I; their transport under change of lattice is governed by Paper I, Theorem 3.2 and Paper I, Theorem 4.2, abstract cells by Paper I, Theorem 4.3, and the failure of laws one might expect by Paper I, Theorem 6.4.

2.5 Factorization

Lemma 2.5 (factorization criterion)**.** *Let $\pi : X \to Y$ be surjective and $f : X \to Z$. Then $f = g \circ \pi$ for some g iff f is constant on every fibre of π; g is then unique. If two points x, x' have $\pi(x) = \pi(x')$ and $f(x) \neq f(x')$, no such g exists.*

Proof. Define $g(y) = f(x)$ for any $x \in \pi^{-1}(y)$; well-definedness is constancy on fibres. The last statement is the contrapositive. □

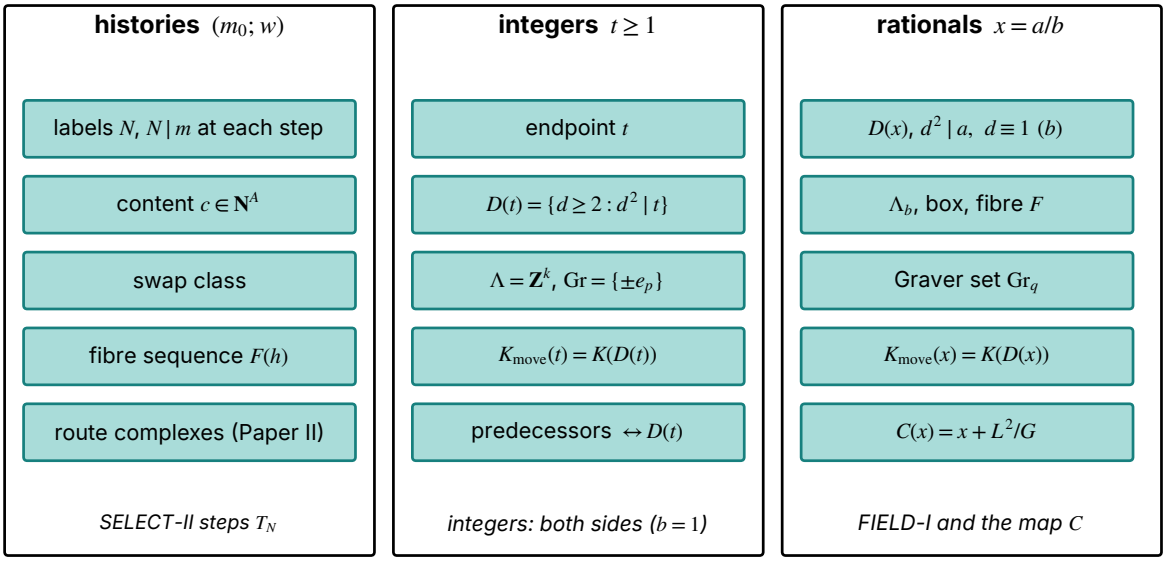


Figure 2: **Arithmetic states and the objects attached to them.** Three domains with their native constructions. Integers are the overlap: an integer is both an endpoint of histories and a collective state with denominator 1. *Schematic of domains.*

Throughout, an *obstruction* to a factorization is a pair x, x' as in the last sentence together with a proof that $\pi(x) = \pi(x')$ and $f(x) \neq f(x')$; its *domain* is the set on which π is taken (a fixed source, or all integers) and is stated each time.
Figure 2 shows arithmetic states and the objects attached to them.

3 Numerical projection and generative information

Fix a source m_0. Let $\mathrm{Hist}(m_0)$ be the set of histories from m_0. Three projections are in play:

$$\mathrm{Hist}(m_0) \xrightarrow{\pi_1} \mathrm{Cls}(m_0) \xrightarrow{\pi_2} \mathrm{Cont}(m_0) \xrightarrow{\pi_3} \mathrm{End}(m_0) \subseteq \mathbb{Z},$$

$$h \mapsto [h] \mapsto c(h) \mapsto m_0 \prod_N R(N)^{c_N} = \mathrm{end}(h).$$

Here $\mathrm{Cls}(m_0)$ is the set of swap classes (all words in a class have the same content, since legal swaps are transpositions) and $\mathrm{Cont}(m_0)$ the set of contents of admissible histories, and $\mathrm{End}(m_0) = \{\mathrm{end}(h)\}$ the set of reachable endpoints; the codomain of π_3 is $\mathrm{End}(m_0)$, not all of $\mathbb{Z}$, so that π_1, π_2, π_3 are surjections. For a finite alphabet A write $\mathrm{Cont}_A(m_0) \subseteq \mathrm{Cont}(m_0)$ for the admissible contents supported on A, $\mathrm{End}_A(m_0)$ for the endpoints of histories over A, and $\pi_{3,A} : \mathrm{Cont}_A(m_0) \to \mathrm{End}_A(m_0)$ for the restriction of π_3; the matrix V_A acts on $\mathbb{Z}^A$ and so only on contents supported on A.

3.1 Cited ingredients

Theorem 3.1 (kernel condition; Paper II, Theorem 9.1)**.** *Let A be a finite alphabet, $\mathcal{P}$ the primes dividing some N or $N+1$, $N \in A$, and $V_A = (\Delta_p(N))_{p\in\mathcal{P},N\in A}$ with $\Delta_p(N) = 2v_p(N+1)-v_p(N)$. Two histories over A from the same source have the same endpoint iff $V_A(c-c') = 0$.*

Theorem 3.2 (faithfulness; Paper II, Theorem 11.3)**.** *For a fixed source the fibre sequence $F(h)$ determines the history h. A legal swap changes exactly one entry of $F(h)$.*

Theorem 3.3 (classification for $\{2,3,14\}$; Paper II, Theorem 5.8)**.** *The number $\nu(m)$ of swap classes of $\{2,3,14\}$ at the source m is 0 if neither 14 nor 21 divides m, 2 if $14 \mid m$ and $\gcd(m/14, 6) = 1$, and 1 otherwise.*

3.2 The proposition

Proposition 3.4 (SIV-P1: endpoint projection and retained information)**.** [NEW, PROVED] *Let $m_0 \geq 1$.*

(a) Factorization. *The maps π_1, π_2, π_3 are well defined and* $\mathrm{end} = \pi_3\pi_2\pi_1$. *Thus the endpoint is a function of the content, and of the swap class, of the history; and a numerical invariant of the endpoint is a function of $(m_0, \mathit{content})$.*

(b) What the endpoint retains. *For histories over a common alphabet A the endpoint determines exactly the vector $V_A c = (v_p(t) - v_p(m_0))_{p\in\mathcal{P}}$, and conversely this vector determines the endpoint. The fibre of $\pi_{3,A}$ over a reachable endpoint $t = \mathrm{end}(h)$ with h over A is $\pi_{3,A}^{-1}(t) = \mathrm{Cont}_A(m_0) \cap \big(c(h) + \ker_{\mathbb{Z}} V_A\big)$: the coset must be intersected with the set of* admissible *contents from m_0 supported on A, not merely with $\mathbb{N}^A$. (Histories using letters outside A and ending at t are in the fibre of π_3 but not of $\pi_{3,A}$; they are compared by enlarging A.) Example: $m_0 = 1$, $A = \{1,2,3\}$, $c = (7,0,0)$ and $c' = (0,2,4)$ have $R(1)^7 = R(2)^2R(3)^4 = 16384$, so $c - c' \in \ker_{\mathbb{Z}} V_A$ and $c' \in \mathbb{N}^A$, yet $c' \notin \mathrm{Cont}_A(1)$ because neither label 2 nor label 3 can be applied first.*

(c) Each projection is non-injective. *(i) π_3: at $m_0 = 32$ the histories $(2,2,2,2)$ and $(1,8,8)$ are admissible with the same endpoint 13122 and different content and length. (ii) π_2: at $m_0 = 14$ the contents of $(14,3,2)$ and $(2,3,14)$ coincide, both end at 5400, but they lie in different swap classes. (iii) π_1: from $m_0 = 1$ the two histories $(1,2,1), (1,1,2)$ to 72 lie in one class.*

(d) Invariants. *Functions of $(m_0, \mathrm{end}(h))$ factor through $\pi_3\pi_2\pi_1$ (this includes, for fixed m_0, the number $H(m_0,t)$ of histories from m_0 to t, which depends only on t); the length r of h, the content and the swap class do not factor through π_3 on any set of sources containing the witnesses in (c). The number of histories to t is a function of t but does* not *factor through the occurrence set $D(t)$ (Proposition 7.1).*

Proof. (a) A legal swap exchanges two adjacent letters, so the content is invariant; hence π_2 is defined on classes. The product formula $\mathrm{end}(h) = m_0 \prod R(N_i)$ gives π_3. (b) is Theorem 3.1 together with the identity $v_p(\mathrm{end}(h)) - v_p(m_0) = \sum_N c_N \Delta_p(N) = (V_A c)_p$, which follows from $v_p(R(N)) = 2v_p(N+1) - v_p(N)$. For the fibre: a content $c' \in \mathrm{Cont}_A(m_0)$ has the endpoint t iff $V_A(c' - c) = 0$, i.e. iff $c' \in c + \ker_{\mathbb{Z}} V_A$; the example has $V_A(c - c') = 0$ by $R(1)^7 = 4^7 = 16384 = (9/2)^2 \cdot (16/3)^4$, and c' is inadmissible from 1 since admissibility of

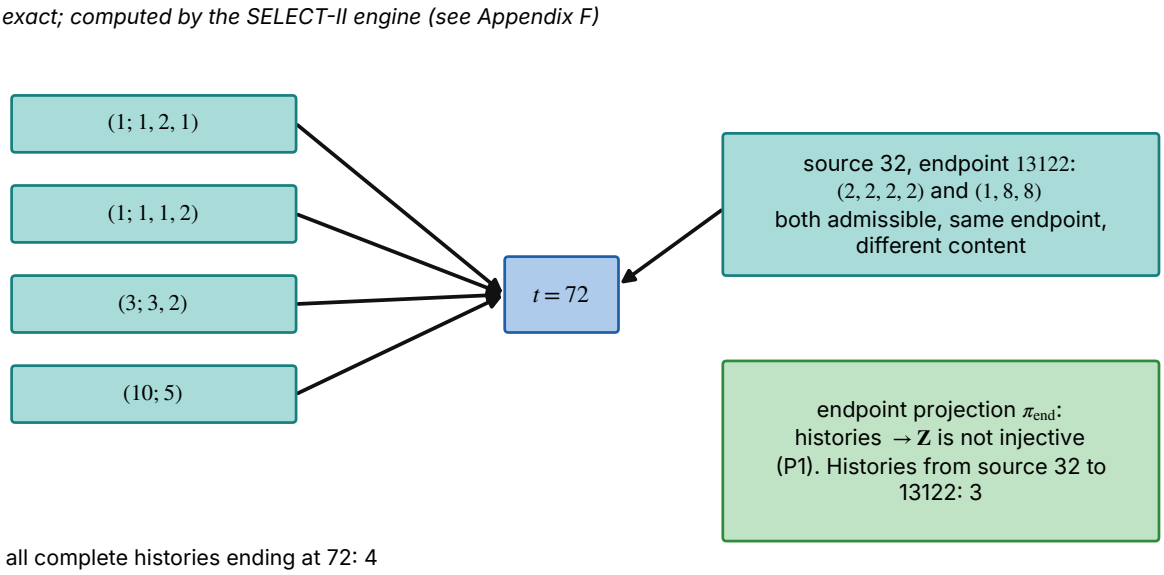


Figure 3: **Endpoint projection and its fibres** (π_3 of Proposition 3.4). Left: the four complete histories that end at 72, from sources $1, 1, 3, 10$. Right: two histories from source 32 with the same endpoint and different content. *Exact; computed by the history engine (Appendix F).*

a first letter N requires $N \mid m_0 = 1$. (c)(i): T_2 applied to $32, 144, 648, 2916$ (all even) gives $144, 648, 2916, 13122$; and $32 \to 128 \to 1296 \to 13122$ by T_1, T_8, T_8 with $8 \mid 128$ and $8 \mid 1296$. Lengths are 4 and 3. (ii): $14 \to 225 \to 1200 \to 5400$ by T_{14}, T_3, T_2 with $3 \mid 225$, $2 \mid 1200$, and $14 \to 63 \to 336 \to 5400$ by T_2, T_3, T_{14} with $3 \mid 63$, $14 \mid 336$. By Theorem 3.3 with $m = 14$ ($14 \mid 14$, $\gcd(1, 6) = 1$) there are two classes; the only admissible arrangements are these two. (iii): $72 = 1 \cdot 4 \cdot \frac{9}{2} \cdot 4$ and the two words differ by one legal adjacent transposition. (d) The first sentence follows from (a) and Lemma 2.5; the non-factorization of r, content and class through π_3 is the witnesses of (c). For the last sentence: the count $H(m_0, t)$ is by definition constant on the fibres of π_3, and its failure to factor through $D(t)$ is the witness pairs of Proposition 7.1. □

Remark 3.5. Proposition 3.4(b) is the precise sense in which the endpoint is a *valuation-displacement*: it forgets how that displacement was composed from letters. The identity $\sum_N c_N \Delta_p(N) = (V_A c)_p$ is the same arithmetic that organises the gain–loss accounting of Paper III, Theorem 13.3.

Figure 3 shows endpoint projection and its fibres.
Figure 4 shows four histories reaching 72 and their structural records.
Figure 5 shows same endpoint, different label content.

4 Occurrence projection and dormant arithmetic structure

The second projection sends a state to its occurrence set. For integers it has a direct generative meaning.

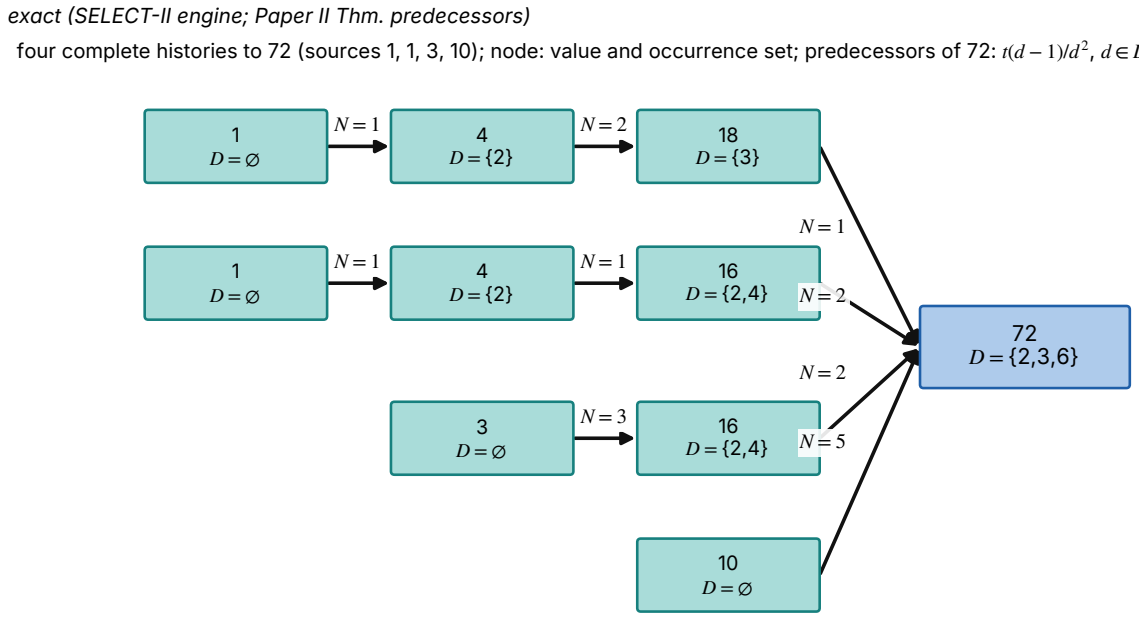


Figure 4: **Four histories reaching 72 and their structural records.** Each node shows an integer and its occurrence set D; arrows are SELECT-II steps labelled by N. The four paths have different fibre sequences $F(h)$, although they share the endpoint 72 and the same terminal data $D(72) = \{2, 3, 6\}$ (Paper II, Theorem 3.1, Paper II, Theorem 11.3). *Exact.*

Proposition 4.1 (SIV-P2: occurrence projection and primitive-relation information)**.** [NEW, PROVED; (B),(C) CITED]

(a) Occurrences are incoming steps. *For an integer $t \geq 1$ the map $d \mapsto (N, m) = \big(d-1,\ t(d-1)/d^2\big)$ is a bijection from $D(t)$ onto the set of generative steps $m \xrightarrow{N} t$. Consequently $|D(t)|$ is the in-degree of t, and for every m_0 the number $H(m_0, t)$ of histories from m_0 to t satisfies $H(m_0, t) = \sum_{d \in D(t)} H\big(m_0,\ t(d-1)/d^2\big)$ (with $H(m_0, m_0) = 1$ and the convention that terms with $t(d-1)/d^2 < m_0$ vanish). Moreover $N + 1 \in D(T_N(m))$ for every admissible step.*

(b) Factorization. *The realised fibre, $K_{\mathrm{move}}(x)$ and all its derived data factor through $x \mapsto D(x)$; for integer t the whole record is a function of $\max D(t)$ (Paper III, Theorem 3.4, Paper III, Theorem 3.1).*

(c) Obstruction. *The primitive relation system does not factor through D: for primes $r \geq 13$, $r \equiv 5 \pmod 8$, the states $x_r = 25r^2/(5r-1)$ and $y_r = 25r^2/8$ have $D(x_r) = D(y_r) = \{5r\}$ and the same active primes and box, but $|\mathrm{Gr}(x_r)| = 2$ and $|\mathrm{Gr}(y_r)| = 4$ (Paper III, Theorem 3.6(i)).*

Proof. (a) If $m \xrightarrow{N} t$ then $t = m(N+1)^2/N$ with $N \mid m$; we have $tN = m(N+1)^2$, so $(N+1)^2 \mid tN$, and $\gcd(N, N+1) = 1$ gives $(N+1)^2 \mid t$, i.e. $d = N+1 \in D(t)$ (here $d \geq 2$), and $m = tN/(N+1)^2 = t(d-1)/d^2$. Conversely if $d \in D(t)$ put $N = d-1$ and $m = t(d-1)/d^2$; then $m \in \mathbb{Z}$ because $d^2 \mid t$, $N \mid m$ because $m = (t/d^2)N$, and $T_N(m) = m(N+1)^2/N = t$. The

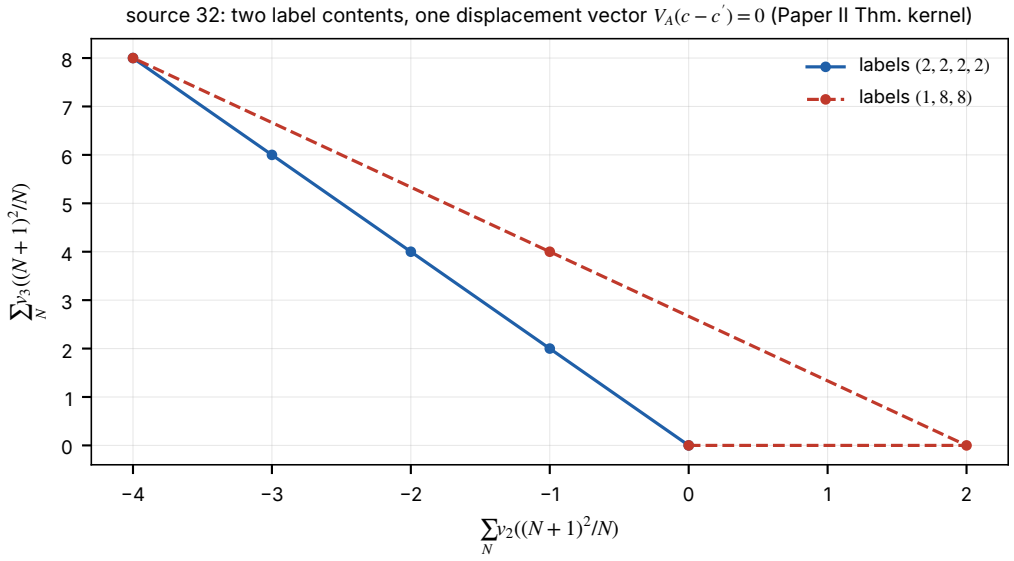


Figure 5: **Same endpoint, different label content.** Cumulative displacement of (v_2, v_3) of the running value for the label words $(2, 2, 2, 2)$ and $(1, 8, 8)$ from source 32: both paths end at $V_A c = V_A c'$ (Theorem 3.1). *Exact.*

two assignments are inverse. A history to t is a history to the predecessor m followed by the step to t, which gives the recursion. For the last statement $(N+1)^2 \mid m(N+1)^2/N = T_N(m)$ because $N \mid m$ makes m/N an integer. (b) and (c) are the cited theorems; for (c) we recomputed $D(x_{13}) = D(y_{13}) = \{65\}$ (Appendix F). □

Corollary 4.2. *The occurrence set of an integer is the branching set of its ancestry: histories to t are sequences of choices of elements of D along the way. Hence the invariant counting histories depends on t through the recursion above, not through $D(t)$ alone (compare Proposition 7.1).*

Information chain. Combining Proposition 4.1 with Proposition 3.4: the chain Hist $\to$ Cls $\to$ Cont $\to \mathbb{Z} \to \{D\} \to \{K_{\text{move}}\} \to \{\beta\}$ consists of surjections, each a factorization of the next by Lemma 2.5; the first three arrows lose information on every set containing the witnesses of Proposition 3.4; the fourth, $t \mapsto D(t)$, is non-injective (all t with the same square-divisor set, Proposition 7.1); and $D \mapsto$ Gr does not exist at all (Proposition 4.1(c)). Figure 20 shows the chain with the status of each arrow.

5 Geometric realization and dimensional transport

5.1 Three tiers of relation

For a state x there are three nested layers of arithmetic relation: the relations of the lattice Λ_b (arithmetic availability), the primitive relations $\text{Gr}_q(x)$, and the *realised moves*, the edges of $K_{\text{move}}(x)$, which are pairs of occurrences present in $D(x)$. Figure 7 depicts the layers; what is proved about each is collected in Proposition 5.3.

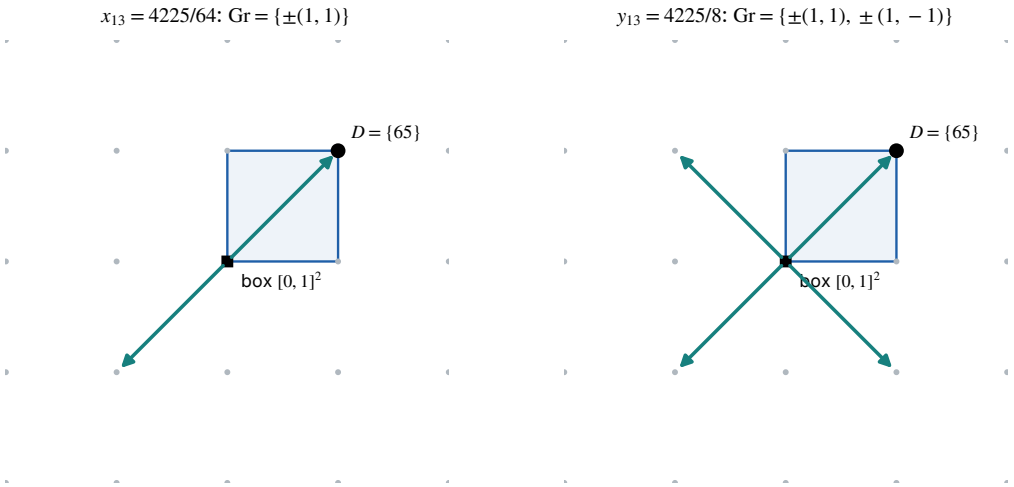


Figure 6: **Occurrence projection and dormant relations.** The states $x_{13} = 4225/64$ and $y_{13} = 4225/8$ have the same occurrence set $\{65\}$, the same active primes $\{5, 13\}$ and the same box $[0, 1]^2$, but the primitive systems differ: the extra pair $\pm(1, -1)$ is a primitive relation of y_{13} that is not realised at x_{13}. *Exact data of Paper III, Theorem 3.6(i) with $r = 13$; D recomputed; the lattice pictures are schematic.*

5.2 An abstract minimality principle

Definition 5.1. Let $(P, \prec)$ be a strict partial order and $X \subseteq P$. Put $\min_{\prec} X = \{g \in X : \nexists\, x \in X,\ x \prec g\}$.

Lemma 5.2 (monotone minimality)**.** *If $X \subseteq X' \subseteq P$ and $g \in \min X$, then $g \in \min X'$ iff no element of $X' \setminus X$ lies below g. In particular, if $X_0 \subseteq X_1 \subseteq \cdots$ and $g \in \min X_{n_0}$, the set $\{n \geq n_0 : g \in \min X_n\}$ is an initial segment of $\{n_0, n_0 + 1, \dots\}$.*

Proof. $g \in X'$. An element of X' below g lies either in X, impossible by minimality in X, or in $X' \setminus X$. For the second claim, if $g \notin \min X_n$ then some $z \in X_n$, $z \prec g$, which stays in $X_{n'}$ for $n' \geq n$. □

Two instances occur in the sources. (I) Paper I: P is a lattice with the conformal order $\sqsubseteq$ on integer vectors, $X = \Lambda \setminus \{0\}$, $\min X = \mathrm{Gr}(\Lambda)$; Paper I, Theorem 4.2 says $g \in \mathrm{Gr}_q(\Lambda')$ iff $g \in \Lambda'$ and no $v \in \Lambda' \setminus \{0, g\}$ satisfies $v \sqsubseteq g$, which is the lemma for $X = \Lambda$, $X' = \Lambda'$ when $\Lambda \subseteq \Lambda'$. (II) Paper III: for a fixed pair $e = \{u, v\}$ with $\gcd(u, v) > 1$ let $P = \{e\} \cup \big(I(u, v) \setminus \{u, v\}\big)$ with $z \prec e$ for every $z \neq e$, and $X_n = \{e\} \cup (D_n \cap I(u, v) \setminus \{u, v\})$. Then $e \in \min X_n$ iff $D_n \cap (I(u, v) \setminus \{u, v\}) = \emptyset$, which by Paper III, Theorem 9.1 is the condition for e to be an edge once $u, v \in D_n$. The two instances share only the logical shape of the criterion; the ambient sets are different.

Proposition 5.3 (SIV-P3: arithmetic availability versus geometric realization)**.** [NEW, PROVED; (C),(D) CITED] *Let $x_n = C^n(x)$ and $D_n = D(x_n)$.*

(a) Availability of vertices is monotone. *If $u, v \in D_n$ then $u, v \in D_{n'}$ for all $n' \geq n$.*

(b) Realisation windows. *For $u, v \in \bigcup_n D_n$, $u \neq v$, let $a(u) = \min\{n : u \in D_n\}$ and $p(u,v) = \min\{n : D_n \cap (I(u,v) \setminus \{u,v\}) \neq \emptyset\}$ (the first time the interval is pierced by an occurrence; ∞ if never; note $1 \in I(u,v)$ only if $\gcd(u,v) = 1$). Then $\{u,v\}$ is an edge of $K_{\mathrm{move}}(x_n)$ iff $\max(a(u), a(v)) \leq n < p(u,v)$ and, when $\gcd(u,v) = 1$, never. In particular the set of times at which a fixed pair is a realised move is an interval of consecutive integers (possibly empty): an edge that disappears never returns.*

(c) Availability does not imply a primitive relation is realised. *For $k \geq 2$ and the family of Paper III, Theorem 4.3 the relation $e_i - e_j$ is primitive at $x = P^2/\ell$ (with $D(x) = \{P\}$) and has no pair of fibre points, hence is not realised; it is realised at $y = (Pq)^2/\ell$.*

(d) Occurrence inclusion does not control primitive persistence *(Paper I, Theorem 6.4, law L1), whereas persistence of a realised move is controlled by D as in (b).*

Proof. (a) is Lemma 2.3. (b) By Theorem 5.4 below, $\{u,v\}$ is an edge of $K_{\mathrm{move}}(x_n)$ iff $u, v \in D_n$ and $I(u,v) \cap (D_n \cup \{1\}) = \{u,v\}$. If $\gcd(u,v) = 1$ then $1 \in I(u,v) \setminus \{u,v\}$ (as $u, v \geq 2$), so there is never an edge. If $\gcd(u,v) > 1$ then $1 \notin I(u,v)$ and the condition is $u, v \in D_n$ and $D_n \cap (I(u,v) \setminus \{u,v\}) = \emptyset$, i.e. $n \geq \max(a(u), a(v))$ and $n < p(u,v)$ by monotonicity of D_n. (c), (d) are cited. □

Theorem 5.4 (divisor-interval rule; Paper III, Theorem 9.1)**.** *Distinct $u, v \in D(x)$ are joined by an edge of $K_{\mathrm{move}}(x)$ iff $I(u,v) \cap (D \cup \{1\}) = \{u,v\}$.*

Corollary 5.5 (answer to S4, structural part)**.** *The persistence criterion for primitive relations in Paper I (Paper I, Theorem 4.2) and the edge-loss criterion in Paper III (Paper III, Theorem 9.1) are instances of Lemma 5.2, with the ambient orders the conformal order on a lattice and the divisor-interval order on D respectively. An implication from one to the other would need an identification of the two ambient sets; this is open (O7) and is* not *claimed.*

5.3 The dimensional boundary of transport

Paper I proves that for lattices $\Lambda \subseteq \Lambda'$ in rank two the realised union of cells is monotone (Paper I, Theorem 7.5) and that this fails in rank three without a free coordinate (Paper I, Theorem 8.2), with a corner-cube criterion (Paper I, Theorem 5.2) and a slab description when a free coordinate exists (Paper I, Theorem 9.1). These statements are about lattice transport, and the relation to occurrence geometry is through Proposition 5.3(c),(d): occurrence inclusion is not a lattice inclusion. Finally, abstract loss of a cell does not mean loss from the realised union (Paper I, Proposition 6.3, specimen $21^8/5 \to 21^8/4$: abstract Betti numbers $(1,5,0) \to (1,2,6)$, realised $(1,2,0) \to (1,0,0)$).
Figure 8 shows planar preservation and three-dimensional failure.
Figure 9 shows abstract cell loss versus realised geometric containment.

6 Arithmetic constraints and history topology

6.1 Local guards and global classes

A single legal swap is a local arithmetic condition: for adjacent letters N, N' at a state m, both $N \mid m,\ N' \mid mR(N)$ and $N' \mid m,\ N \mid mR(N')$ must hold. Whether the *global* space of

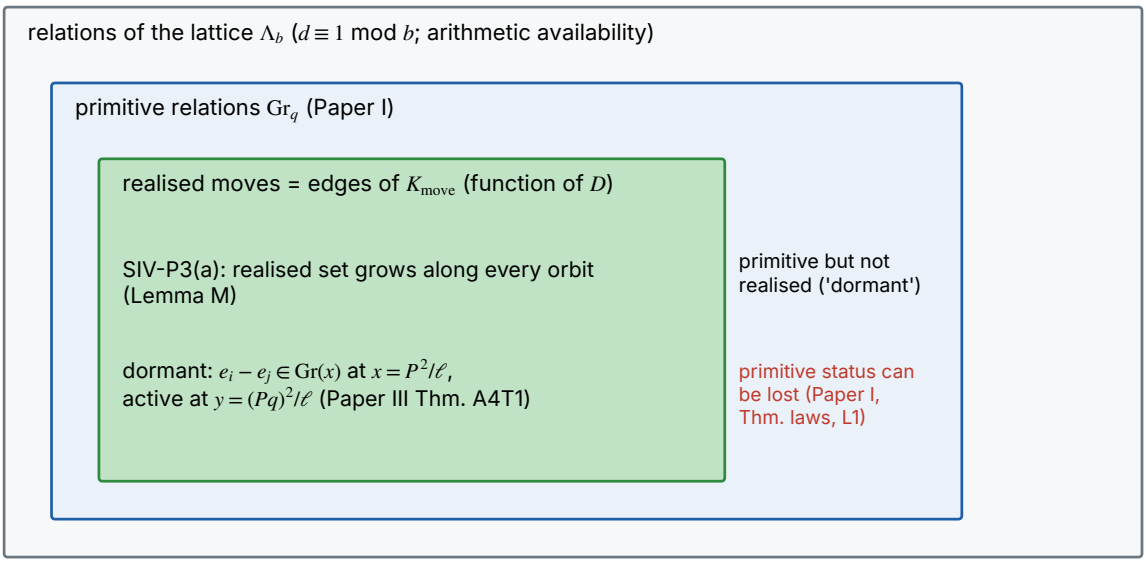


Figure 7: **Arithmetic possibility versus geometric realization.** The relation lattice contains the primitive relations, which contain what is realised as moves of K_{move}; the realised set is a function of D and grows along orbits (Proposition 5.3(a)), primitive status can be lost. *Schematic of containments; cited exact examples are named.*

arrangements of a content is connected depends on how these conditions interact along all positions. The sharpest exact statement available is the classification of $\{2, 3, 14\}$ (Theorem 3.3), together with a general separation principle.

Theorem 6.1 (a separator of swap classes; Paper II, Theorem 11.9)**.** *Let A be a set of distinct labels with $A \cap \{1, 2, 4\} = \emptyset$, the source 4 and the content $\{2, 4\} \cup A$. For an admissible arrangement w let $I(w) = 1$ if 2 precedes 4 and 0 otherwise. Then I is constant on swap classes; consequently the number of classes is at least the number of values of I that occur.*

Theorem 6.2 (route cubes; Paper II, Theorem 8.2)**.** *For every $r \geq 1$ the complete histories ending at 72^r contain an r-cube: from source 1, the prefix 1^r followed by r blocks, each independently $(1, 2)$ or $(2, 1)$.*

Proposition 6.3 (SIV-P4: arithmetic constraints and global history connectivity)**.** [NEW, PROVED FROM CITED THEOREMS; FINITE CHECK]

(a) Swap-invariant separators. *If ϕ is a function on the admissible arrangements of a content that is invariant under legal swaps, the number of swap classes is at least $|\phi(\text{arrangements})|$. (An obstruction to connectivity is thus certified by one invariant.)*

(b) Connectivity is not monotone in the source. *For $\{2, 3, 14\}$: $\nu(7) = 0$, $\nu(14) = 2$, $\nu(28) = 1$, $\nu(70) = 2$. Hence the class count increases from 7 to 14, decreases from 14 to 28, and is not a function of divisibility order.*

(c) Connectivity is not a function of content and displacement alone. *The content $\{2, 3, 14\}$ gives $\nu = 2$ at $m = 14$ and $\nu = 1$ at $m = 28$; the displacement vector $V_A c$ is the same in both*

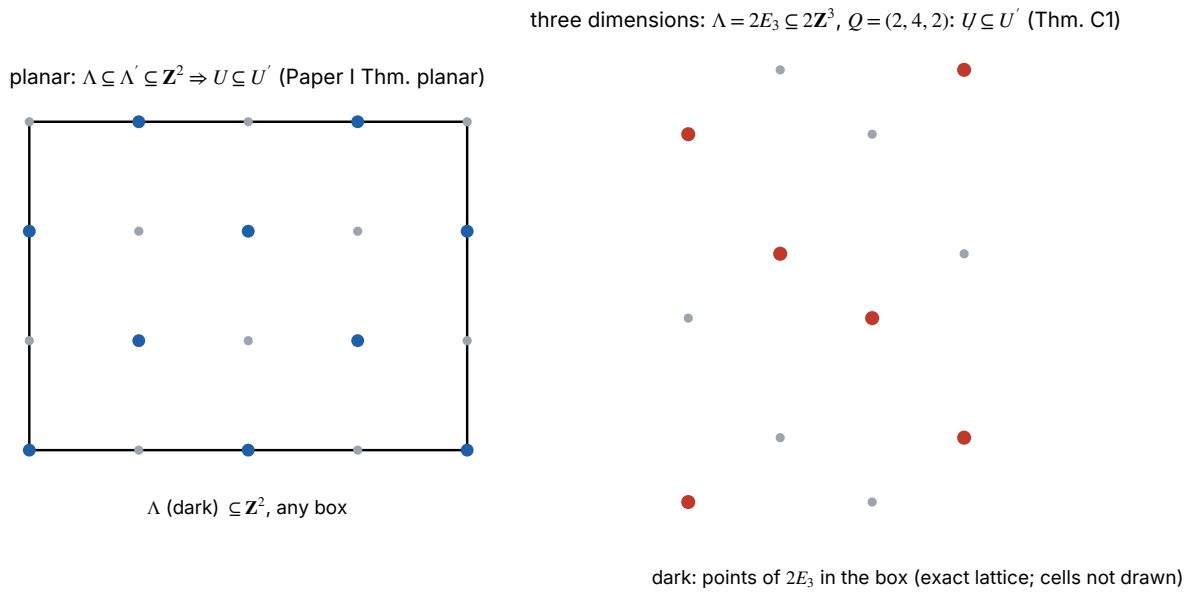


Figure 8: **Planar preservation and three-dimensional failure.** Left: Paper I, Theorem 7.5. Right: Paper I, Theorem 8.2 with $\Lambda = 2E_3 \subseteq 2\mathbf{Z}^3$ and box $(2,4,2)$. *Theorem-level schematic: lattice points are exact, cells are not drawn.*

cases (it depends on the content only). Hence content together with valuation displacement, without the source, does not determine the class count across different sources. (This compares two sources and is therefore not a statement about the information retained by the endpoint projection π_3 of one fixed source; for that see Proposition 3.4(b) and Proposition 7.1.)

(d) Local freedom with global rigidity. *The 2-cube of histories* $1 \to 72^2$ *consists of four histories joined in a 4-cycle of legal swaps; it is a subgraph of the swap graph of all* 16 *histories* $1 \to 5184$ *(exhaustive count).*

Proof. (a) Constancy on classes plus distinct values give distinct classes. (b) Substituting into Theorem 3.3: $m = 7$: neither 14 nor 21 divides 7, so $\nu = 0$; $m = 14$: $14 \mid 14$, $\gcd(1,6) = 1$, so $\nu = 2$; $m = 28$: $14 \mid 28$, $\gcd(2,6) = 2$, so $\nu = 1$; $m = 70$: $\gcd(5,6) = 1$, $\nu = 2$. Since $7 \mid 14 \mid 28$ and $14 \mid 70$ the pattern up, down, up shows non-monotonicity in the divisibility order. (A direct check of the formula on all sources $1 \le m < 1500$ found no deviation; Appendix F.) (c) is read off from (b): $\nu(14) \neq \nu(28)$ with the same content, hence the same $V_A c$, at the two different sources 14 and 28. (d) is Theorem 6.2 for $r = 2$ together with the exhaustive history count of Appendix F; adjacent words in the cube differ by one transposition of a block, which is legal since both orders are histories. □

Figure 10 shows local swaps and globally disconnected history classes.

6.2 Cubes and homology

A combinatorial cube of legal swaps is contractible once filled; the homology of the route complex of all histories to an integer records more. Paper II proves that route complexes can

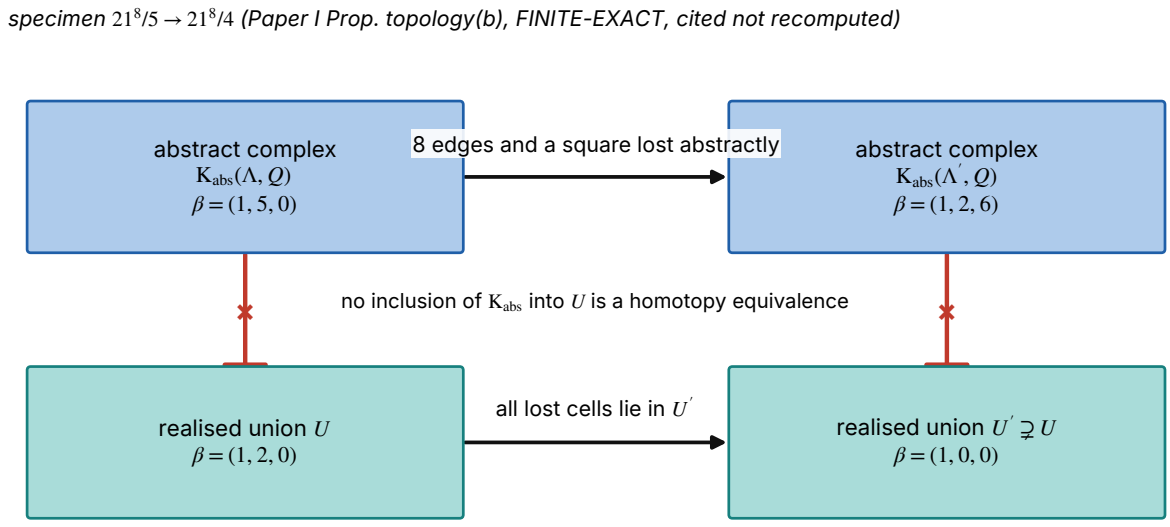


Figure 9: **Abstract cell loss versus realised geometric containment.** For the planar specimen of Paper I, Proposition 6.3(b) the abstract complex loses cells while the realised union grows. *Finite-exact values quoted from Paper I, not recomputed.*

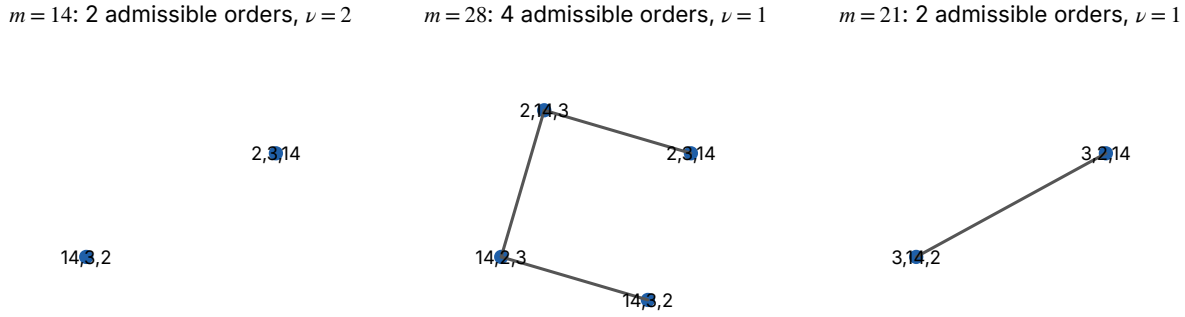


Figure 10: **Local swaps and globally disconnected history classes.** Swap graphs of the admissible orders of $\{2, 3, 14\}$ at the sources 14, 28, 21 (Paper II, Theorem 5.8); the number of components is $\nu = 2, 1, 1$. *Exact; computed.*

have nontrivial higher-dimensional homology in suitable families (see Paper II, Theorem 7.5, Paper II, Theorem 7.3); we use only the logical point that *the presence of cubes does not decide the homology*, which is why the history side requires its own computations and cannot be read from $K(D(t))$ (Proposition 7.1).
Figure 11 shows a history loop and a higher-dimensional void.

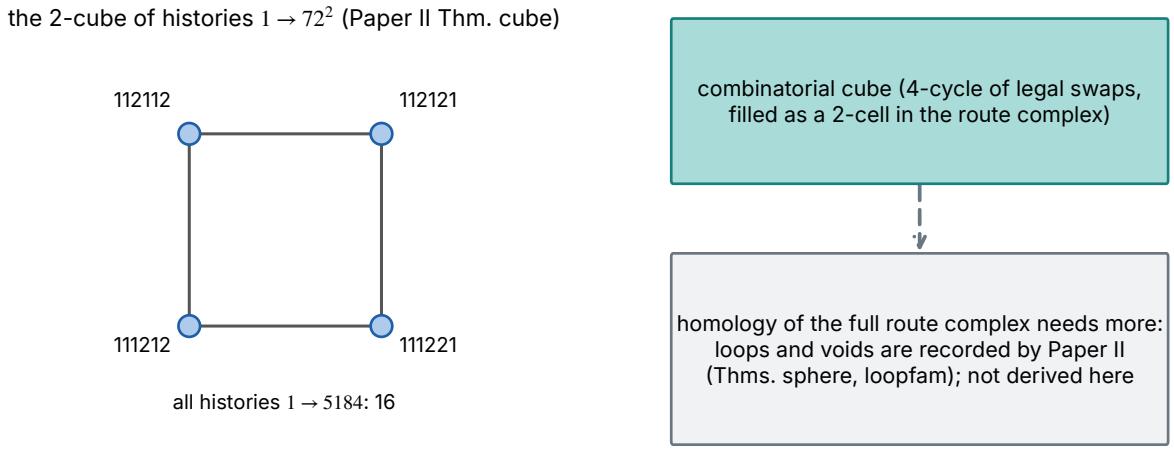


Figure 11: **A history loop and a higher-dimensional void.** Left: the 2-cube of histories $1 \to 5184$ (exact words). Right: what is and is not claimed; the homology of full route complexes is recorded in Paper II. *Left exact, right schematic.*

7 Endpoint geometry versus global ancestry

For an integer t the endpoint geometry is $K(D(t))$ together with everything determined by $D(t)$ (Proposition 4.1). The ancestry of t is its set of complete histories, their classes and their complexes. Proposition 7.1 separates them.

Proposition 7.1 (SIV-P5: limits of endpoint geometry as a history invariant)**.** [NEW, PROVED BY EXHAUSTIVE RECURSION; FINITE-EXACT WITNESSES]

(a) Obstruction. $D(900) = D(9000) = \{2, 3, 5, 6, 10, 15, 30\}$, *hence* $K(D(900)) = K(D(9000))$ *and all of the data of Proposition 4.1(b) coincide, but from the source* 4 *there are* 2 *histories forming* 1 *swap class to* 900 *and* 4 *histories forming* 2 *swap classes to* 9000. *Likewise* $D(288) = D(4320)$ *with* (3 *histories*, 1 *class*) *and* (3, 2) *from source* 4.

(b) Consequence. *Neither the number of histories, nor the number of swap classes, nor the isomorphism type of the swap graph of* $\mathrm{Hist}(4, t)$ *factors through* $t \mapsto D(t)$ *(Lemma 2.5);* a fortiori *none of them factors through* $K(D(t))$ *or its Betti numbers.*

(c) Frequency (finite-exact). *Let* E *range over the integers* $16 \le E \le 120000$ *that are not squarefree and are reachable from the source* 4. *They realise* 238 *distinct occurrence sets* $D(E)$*;* 166 *of these sets are shared by several* E*, and for* 106 *of them the number of swap classes is not constant (for* 158 *the number of histories is not constant). This is a finite census and is not extrapolated.*

(d) What would suffice. *The fibre sequence* $F(h)$ *determines the history (Theorem 3.2); so the occurrence sets* along the path *reconstruct ancestry, but the occurrence set of the endpoint does not (S2, S3).*

Proof. (a) By Proposition 4.1(a) the history counts satisfy $H(4, t) = \sum_{d \in D(t)} H(4, t(d-1)/d^2)$; the recursion terminates because predecessors decrease. We evaluated it exactly for 900 and

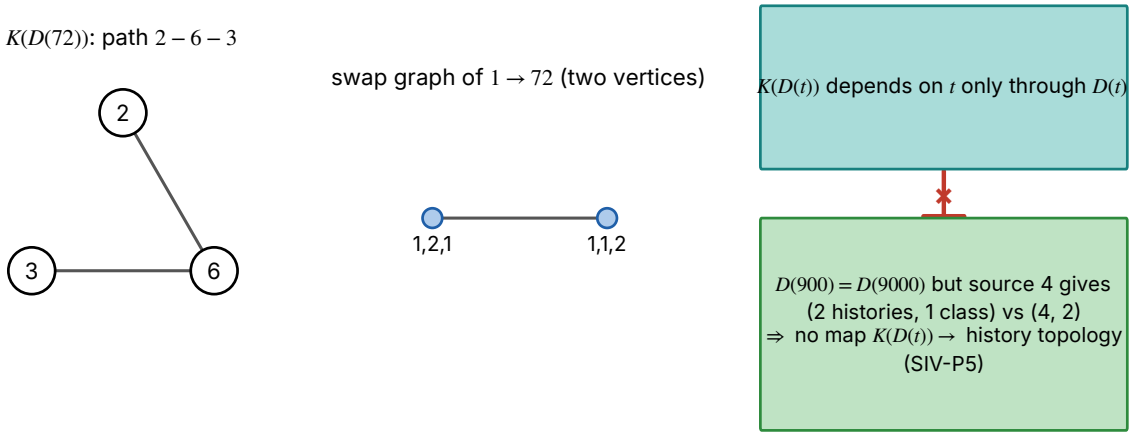


Figure 12: **Endpoint move complex versus history complex.** Left: $K(D(72))$ is the path $2-6-3$. Middle: the swap graph of the histories $1 \to 72$ has two vertices. Right: the obstruction SIV-P5. *Exact; the right panel is schematic.*

9000 (and for $288, 4320$); the 2 histories to 900 and the 4 to 9000 are listed in Appendix E, and the swap classes are computed by union–find over legal adjacent transpositions of those words. $D(900) = D(9000)$ is read off from $900 = 2^2 3^2 5^2$ and $9000 = 2^3 3^2 5^3$: in both, d^2 divides the number iff $d \mid 30$. (b) is Lemma 2.5. (c) is a recorded exhaustive computation (Appendix F). (d) is Theorem 3.2. □

Figure 12 shows endpoint move complex versus history complex.

8 Structural changes during collective evolution

Let x_0 be a state with $D(x_0) \neq \emptyset$ and $x_n = C^n(x_0)$, $D_n = D(x_n)$, $K_n = K(D_n)$. By Lemma 2.3 $D_0 \subseteq D_1 \subseteq \cdots$, but the complexes K_n are not nested: new occurrences pierce intervals and kill edges. The question is which structural features persist.

8.1 Connectivity is governed by prime supports

Lemma 8.1 (component rule)**.** *For a finite $D \subset \mathbb{Z}_{\geq 2}$ the components of $K(D)$ are the classes of the equivalence relation on D generated by $u \approx v \iff \gcd(u, v) > 1$. Equivalently $\beta_0(K(D))$ is the number of connected components of the bipartite graph between D and the set of primes dividing elements of D (with $d \sim p$ iff $p \mid d$).*

The proof is in Appendix C; it uses only the interval criterion Theorem 5.4: a coprime pair is never an edge because $1 \in I(u, v)$, and a pair with $\gcd > 1$ is joined by a path obtained by induction on $|I(u, v)|$. The lemma was also checked on 3000 random sets (Appendix F).

Proposition 8.2 (SIV-P6: hierarchy of structural persistence under evolution)**.** [NEW, PROVED; (D) USES CITED CENSUS] *Along an orbit:*

(a) Vertices *persist:* $D_n \subseteq D_{n+1}$.

(b) Connectivity *persists: if $u, v \in D_n$ lie in one component of K_n they lie in one component of K_{n+1}. Hence each component of K_n is contained in a unique component of K_{n+1}, and β_0 changes only by merging old components and by adding components consisting of new vertices; a component never splits.*

(c) Edges *do not persist in general: a pair $\{u, v\}$ is an edge during exactly the interval $[\max(a(u), a(v)), p(u, v))$ of Proposition 5.3(b). In particular an edge that is lost is never regained.*

(d) Degrees 0 and 1 are not monotone. *For arbitrary inclusions of finite sets $D \subseteq D' = D \cup N$ of integers ≥ 2 (abstract transitions, not claimed to be occurrence inventories of rational states or steps of an orbit) there are exact witnesses for both directions of change in each of the two degrees: $\Delta\beta_0 = -1$ for $D = \{3, 10\}$, $N = \{18\}$ ($\beta_0 : 2 \to 1$) and $\Delta\beta_0 = +1$ for $D = \{6, 9\}$, $N = \{3, 5\}$ ($1 \to 2$); $\Delta\beta_1 = +1$ for $D = \{8, 12\}$, $N = \{2\}$ ($\beta_1 : 0 \to 1$) and $\Delta\beta_1 = -1$ for $D = \{5, 14, 33, 60, 90\}$, $N = \{6\}$ ($\beta_1 : 3 \to 2$). On realised orbits, the census of Paper III, Theorem 11.6 over 2,576 occurrence events has β_1 decreasing in 176 and increasing in 388, so degree 1 is non-monotone along the examined orbits. Orbit-level non-monotonicity of β_0 is* not *established by the abstract witnesses: for instance $D = \{3, 10\}$ is not the occurrence set $D(x)$ of any state x, since $3, 10 \in D(x)$ force the denominator b to divide both 2 and 9, hence $b = 1$, and then $10 \in D(x)$ forces $2 \in D(x)$. For degree 2 the census records changes in 246 events and for degree 3 in one event; no direction of change is claimed there, and monotonicity of β_2, β_3 is not decided by this paper.*

In particular the persistence is strictly graded: vertices $\supsetneq$ connectivity relation $\supsetneq$ edges, with homology of positive degree unconstrained beyond Paper III, Theorem 10.4.

Proof. (a) is Lemma 2.3. (b) By Lemma 8.1 u, v are in one component of K_n iff they are joined by a chain $u = w_0, \ldots, w_s = v$ in D_n with $\gcd(w_{i-1}, w_i) > 1$. The same chain lies in $D_{n+1} \supseteq D_n$. Components of K_{n+1} either contain an old component or consist of new vertices, so $\beta_0(K_{n+1}) = \#\{\text{classes of } D_{n+1}\}$ counts merged old classes once and new classes. (c) is Proposition 5.3(b). (d) The four set-level witnesses are recomputed exactly (Appendix F, witness W9, abstract set level); for instance $\{3, 10\}$ has two components (they share no prime) and 18 shares primes with both, giving $\beta_0 = 2 \to 1$, while $\{6, 9\} \cup \{3, 5\}$ adds the isolated vertex 5, giving $1 \to 2$. The events of Figure 15 are further instances for degree 0. The census statements are quoted from Paper III, Theorem 11.6(b); they are finite-exact, are not recomputed here, and are used only as existence statements. Strictness of the grading is witnessed by the same events together with (c) (an edge between old vertices is lost while the connectivity relation is unchanged, Figure 14). □

Remark 8.3. Proposition 8.2(b) is a statement about the relation $\sim_n$ on D_n; it does not say that β_0 is monotone. It says that all non-monotonicity of β_0 comes from the two sources named.

Figure 13 shows dormant-to-active structure along an actual collective trajectory.

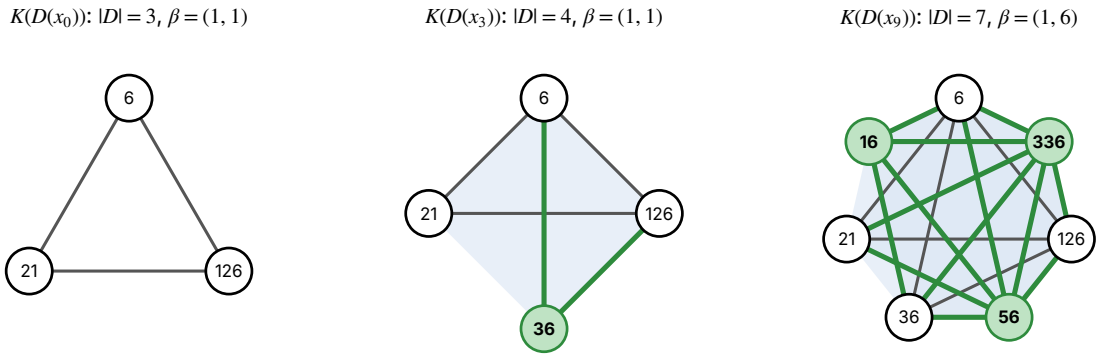


Figure 13: **Dormant-to-active structure along an actual collective trajectory.** Complexes $K(D(x_n))$ at $n = 0, 3, 9$ of the orbit of $x_0 = 126^2/5$; green marks occurrences and moves that are new relative to the previous panel. The occurrence set grows, edges are born at arrival times. *Exact complexes; vertex placement on a circle carries no meaning.*

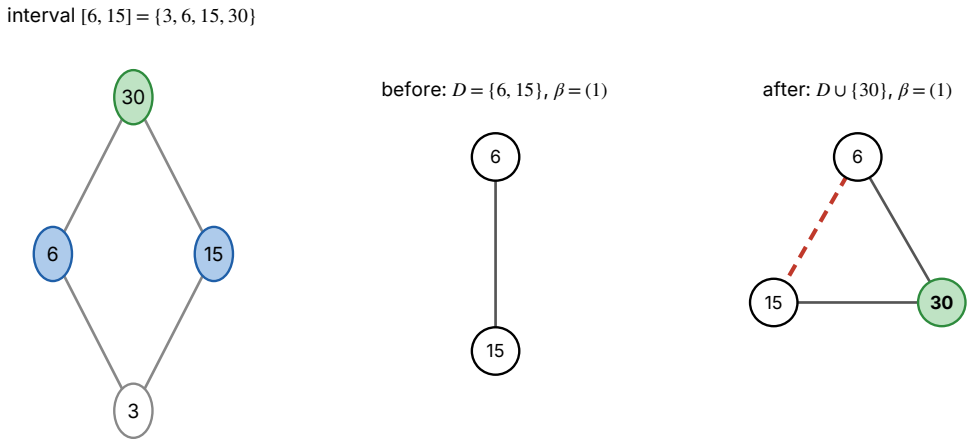


Figure 14: **Occurrence persistence with primitive-edge loss.** The divisor interval $I(6, 15) = \{3, 6, 15, 30\}$ and the complexes $K(\{6, 15\})$ and $K(\{6, 15, 30\})$: the new occurrence 30 pierces the interval and the edge 6–15 is lost (dashed), replaced by 6–30 and 15–30 (Paper III, Theorem 9.1). *Exact.*

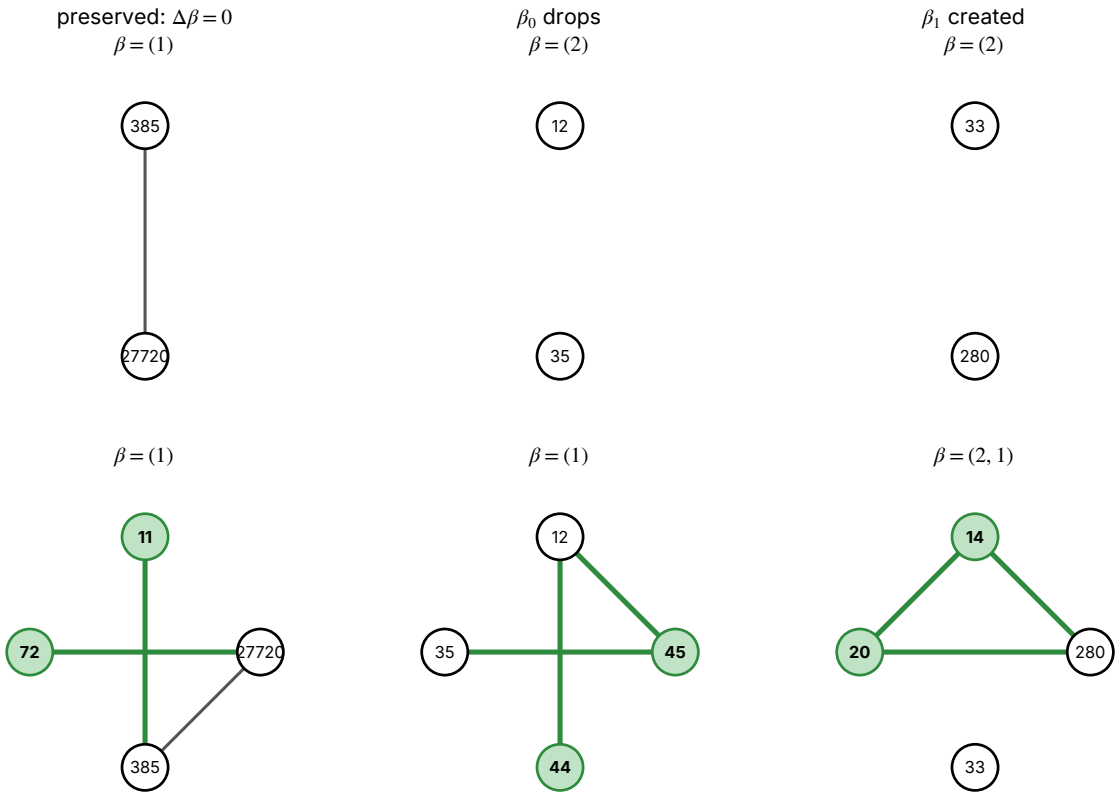


Figure 15: **Topological creation, preservation and destruction.** Three exact events $K(D) \to K(D \cup N)$: unchanged Betti numbers, loss of a component, creation of a first homology class. *Exact complexes computed by the engine of Paper III; β over $\mathbb{F}_2$.*

9 Arithmetic prediction and homological consequences

9.1 What arithmetic predicts

Paper III proves that the step at which the occurrence set next changes is given by a congruence sieve: with $Y_j = ah + jL^2$ the integers $d \geq 2$ occur at the stage j iff $Q(d) \mid Y_j$ (Paper III, Theorem 7.9), so the event time is computed from divisibility alone (Paper III, Corollary 7.10); and that the first occurrences that arrive in an activation family are determined by explicit moduli (Paper III, Theorem 8.5). The homological consequences are governed by the zig-zag identity $K \leftarrow K^\circ \to K'$ (Paper III, Theorem 10.1) and, for nested events, $\Delta\beta_i = h_i(K', K) - k_i - k_{i-1}$ (Paper III, Theorem 10.6). The question is how much of $\Delta\beta$ is determined by cheaper data.

Figure 16 shows arithmetic prediction of collective occurrence events.

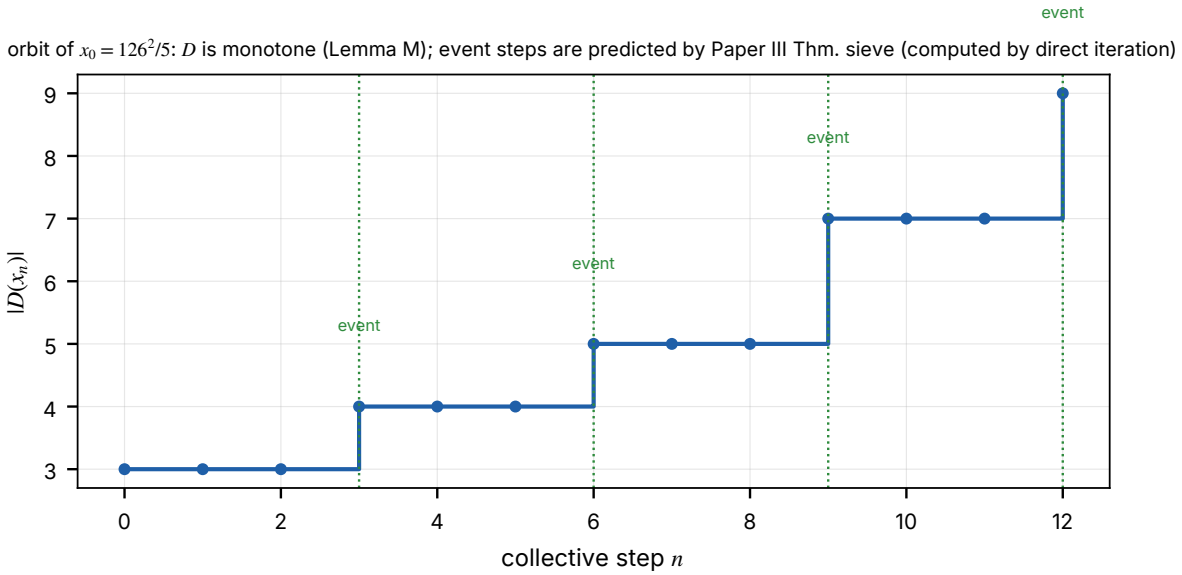


Figure 16: **Arithmetic prediction of collective occurrence events.** The size $|D(x_n)|$ along the orbit of $126^2/5$: monotone with events at the steps predicted by the stage sieve (Paper III, Theorem 7.9). *The mechanism is the cited theorem; the plotted values are computed by direct iteration.*

9.2 Information requirements

Proposition 9.1 (SIV-P7: information requirements for topological prediction)**.** [NEW, PROVED; WITNESSES FINITE-EXACT] *Let $D \subseteq D' = D \cup N$ be finite sets of integers ≥ 2 (statements (a)–(d) concern the abstract complexes $K(D), K(D')$; the witnesses of (d) are abstract and are not claimed to occur along collective orbits) with $K = K(D)$, $K' = K(D')$, local data $\lambda = (f(K), f(K'),$ dead cells, new cells$)$, where f is the cell-count vector.*

(a) Euler. $\sum_i (-1)^i \Delta\beta_i = \sum_i (-1)^i \Delta f_i$*; the right side is a function of* λ*.*

(b) First Betti number from supports. $\Delta\beta_0 = \mathrm{cl}(D') - \mathrm{cl}(D)$ *where* cl *is the number of classes of the share-a-prime relation (Lemma 8.1); it is a function of the prime supports of D and N and does not need the complexes.*

(c) Graph level is determined. *If K and K' have no 2-cells, then $\Delta\beta_1 = \Delta f_1 - \Delta f_0 + \Delta\beta_0$ and $\Delta\beta_i = 0$ for $i \geq 2$. Thus $(\lambda,$ supports$)$ determine $\Delta\beta$ completely.*

(d) Obstruction: λ alone does not determine $\Delta\beta$ (abstract set-level events). *The events $D = \{385, 27720\}$, $N = \{11, 72\}$ and $D = \{15, 315\}$, $N = \{4, 21\}$ have the same λ ($f = (2,1,0) \to (4,3,0)$, nothing lost, 2 new vertices and 2 new edges), with $\Delta\beta = (0,0,0)$ and $(+1,+1,0)$. Likewise $D = \{12, 35\}, N = \{44, 45\}$ and $D = \{33, 280\}, N = \{14, 20\}$ have $f = (2,0,0) \to (4,3,0)$ and $\Delta\beta = (-1,0,0)$ versus $(0,+1,0)$.*

(e) Beyond graph level (open). *When 2-cells are present, λ together with $\Delta\beta_0$ fixes only the alternating sum $\Delta\beta_1 - \Delta\beta_2 + \cdots$. We know of no pair with equal λ and equal $\Delta\beta_0$ but different $(\Delta\beta_1, \Delta\beta_2)$: a random search of* 60,000 *events (*2,039 *distinct values of $(\lambda, \Delta\beta_0)$ with 2-cells) found none. This is not a theorem (O4).*

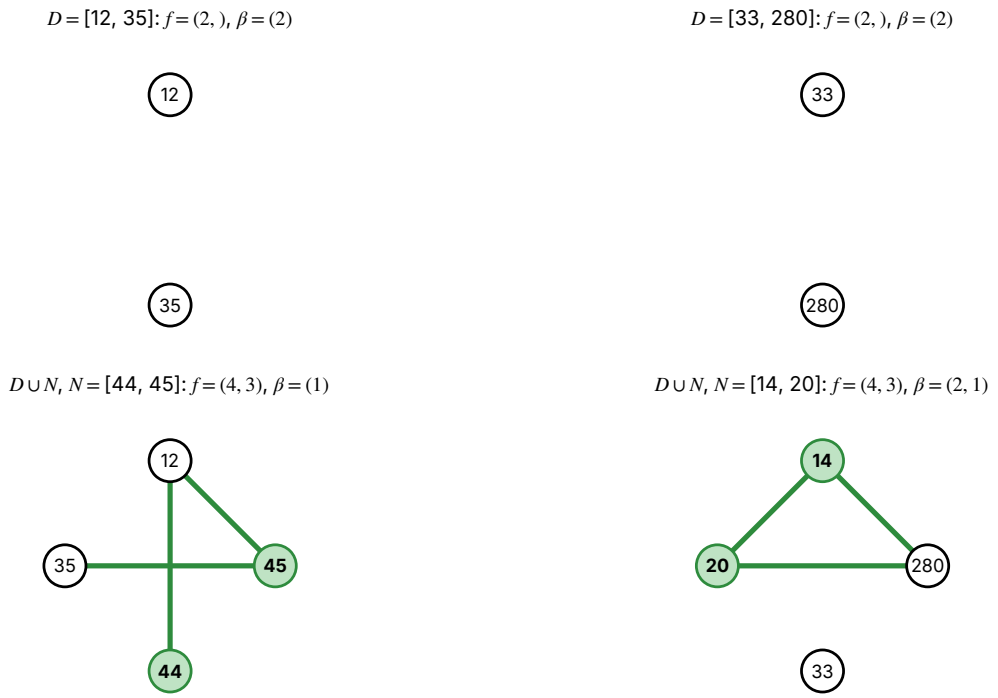


Figure 17: **Surviving complex and homological attachment information.** Two events with identical local counts $f = (2,0,0) \to (4,3,0)$ and different Betti changes: $\Delta\beta = (-1,0,0)$ and $(0,+1,0)$. The difference is carried by the prime supports of the new vertices relative to the old components (Proposition 9.1(b),(c)). *Exact.*

Proof. (a) For finite complexes $\chi = \sum(-1)^i\beta_i = \sum(-1)^i f_i$ over any field. (b) is Lemma 8.1 applied to D and D'. (c) By (a) with $f_i = 0$ for $i \geq 2$: $\Delta\beta_0 - \Delta\beta_1 = \Delta f_0 - \Delta f_1$, and higher Betti numbers vanish for a graph. (d) The values of λ and the Betti numbers of the four complexes are recomputed with the exact engine (Appendix E, E5); they also illustrate (b) and (c): in the first pair $\Delta f_1 - \Delta f_0 = 0$, so $\Delta\beta_1 = \Delta\beta_0$, which is 0 for $\mathrm{cl} : 1 \to 1$ and $+1$ for $\mathrm{cl} : 1 \to 2$; in the second pair $\Delta f_1 - \Delta f_0 = +1$ and $\mathrm{cl} : 2 \to 1,\ 2 \to 2$ give $\Delta\beta_1 = 0$ and $+1$. (e) records a search outcome. □

Remark 9.2. The arithmetic therefore predicts *when* the occurrence set changes (sieve) and *what* β_0 and, at graph level, β_1 become (supports and local counts). What is not predicted from local counts is how new cells attach to old homology classes in positive degree; that information is the attachment data $(r_i, h_i, \beta(K^\circ))$ of Paper III, Theorem 10.1.

Figure 17 shows surviving complex and homological attachment information.

10 Collective recurrence and geometric behavior

Paper III studies two infinite families of towers in which the orbit returns, after a fixed number of steps, to a multiple of its starting point: for $G = 5$ the period is 3 steps with multiplier 4 (Paper III, Theorem 6.2), and for $G = 11$ it is 8 steps with multiplier 9 (Paper III, Theorem 6.1). The Betti numbers of the corresponding complexes are known exactly for $k \leq 140$ (Paper III, Theorem 12.10) and eventually quasi-polynomial without an effective threshold (Paper III, Theorem 12.4). The *scaling data* of a tower are the period, the multiplier, the value of G and the growth law of L_k. The question here is whether the scaling data determine the topology.

Proposition 10.1 (SIV-P8: scaling information versus geometric structure)**.** [NEW; PROVED BY EXACT COMPUTATION ON STATED RANGES (FINITE-EXACT)] *For $S_0 \in \{126, 3094\}$ and $x_k = (S_0 2^k)^2/5$ one has $C^3(x_k) = 4x_k = x_{k+1}$, $L(x_k) = S_0 2^k$ and $G(x_k) = 5$ for $0 \leq k \leq 11$ (twelve periods); the two towers therefore have the same* normalized scaling signature *$\sigma = (G, \text{period}, \text{multiplier}, \rho) = (5, 3, 4, 2)$, where $\rho = L(x_{k+1})/L(x_k)$ is the growth ratio of L. (The towers do* not *share the values of L itself: $L(x_0) = 126$ and 3094 respectively.) Their complexes $K(D(x_k))$ have, for each of $k = 11, \ldots, 14$, Betti numbers $(1, 6, 5, 0)$ for $S_0 = 126$ and $(1, 10, 9, 1, 0)$ for $S_0 = 3094$ (over $\mathbb{F}_2$). Hence the parameters G, the period, the multiplier and the growth ratio ρ do not determine the Betti vector, even for k in the stable range; in particular β_3 and the total Betti sum differ. No claim is made about any other scaling parameter, in particular about the initial value $L(x_0)$ or its divisor lattice.*

Proof. Direct exact computation (Appendix F): iterating C on rationals for 36 steps from x_0, comparing $C^{3j+3}(x_0)$ with $4C^{3j}(x_0)$ and computing L, G; and computing $K(D)$ and its $\mathbb{F}_2$-Betti numbers for $S_0 2^k$, $0 \leq k \leq 14$. For $S_0 = 126$ the values $(1, 6, 5, 0)$ at $7 \leq k \leq 140$ are proved finite-exact in Paper III, Theorem 12.10; our computation agrees on $7 \leq k \leq 14$. The statement concerns only the stated ranges and makes no claim about larger k. □

Remark 10.2. The seeds are $126 = 2 \cdot 3^2 \cdot 7$ and $3094 = 2 \cdot 7 \cdot 13 \cdot 17$. The extra prime factors of 3094 coincide with a non-zero β_3 although the modulus 5, the period and the multiplier are identical. We do not identify which property of the divisor lattice of S_0 controls the Betti vector; that is the content of the open problem O5.

Figure 18 shows $G = 5$ scaling versus geometric evolution.

11 Correspondence between distinct generating rules

The two rules are distinct transformations; the question is at which levels a precise relation exists. We separate five levels.

Definition 11.1 (correspondence levels)**.** **L1** numerical ratios (values of $C^n(x)/x$ and $T_N(x)/x = R(N)$); **L2** occurrence information (the sets D and their transport); **L3** reduced event records (which labels or occurrences appear when); **L4** geometric structures (complexes, Betti numbers); **L5** dynamical intertwining ($\Phi \circ C = T \circ \Phi$ for a map Φ between the state spaces).

Theorem 11.2 (cited: individual versus collective dynamics)**.** *(i) For every state with $D(x) \neq \emptyset$: $1 < C(x)/x \leq 2$, whereas $R(N) \geq 4$; hence $C(x) \neq T_N(x)$ for all N (Paper III, Proposition 13.6). (ii) On the $G = 5$ towers $C^3(x) = 4x = T_1(x)$; $R(N) \in \mathbb{Z}$ iff $N = 1$; the multiplier*

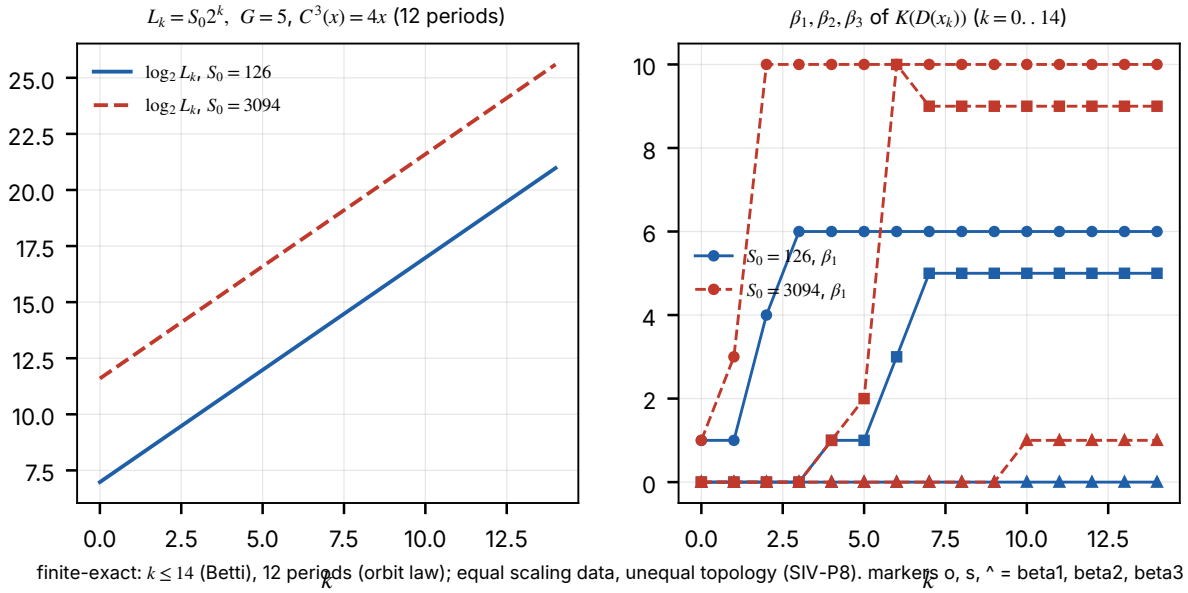


Figure 18: $G = 5$ **scaling versus geometric evolution.** Two towers with the same scaling signature $(G, \text{period}, \text{multiplier}, \rho) = (5, 3, 4, 2)$ but different $L(x_0)$ (left) and different Betti numbers (right). *Finite-exact: orbit law verified for twelve periods, Betti numbers for* $0 \le k \le 14$*; no extrapolation.*

9 *of the* $G = 11$ *towers is not* $R(N)$*; and all coincidences* $x_m/x_n = R(N)$ *in the towers are classified in Paper III, Theorem 13.12.*

Proposition 11.3 (SIV-P9: correspondence levels for distinct arithmetic dynamics)**.** [NEW FOR (C); ASSEMBLED FROM CITED RESULTS FOR (A),(B),(D)]

(a) One-step identity correspondence is obstructed. *There is no* N *with* $C = T_N$ *on any state with* $D \neq \emptyset$*; consequently the identity map* $\Phi = \mathrm{id}$ *(or the inclusion* $\iota : \mathbb{Z} \to \mathbb{Q}_{>0}$*) does not intertwine one step of* C *with one step of any* T_N*. This excludes only the identity-type correspondence; it does* not *exclude a non-identity* Φ *with* $\Phi \circ C = T \circ \Phi$*, and the existence of such a* Φ *(the general L5 problem) is open (O3). The snapshot relation* $C^3 = T_1$ *on the towers is likewise not a conjugacy (Paper III, Theorem 13.7).*

(b) L1 has a rigid, restricted positive result. $C^3 = T_1$ *on the* $G = 5$ *towers; for* $G = 11$ *and the multiplier* 9 *there is no ratio coincidence; the full list of coincidences is Paper III, Theorem 13.12.*

(c) L2 has a proved correspondence on integer states. *Let* $x \in \mathbb{Z}_{\ge 1}$ *with* $D(x) \neq \emptyset$ *and* $C(x) \in \mathbb{Z}$*. Then* $D(x) \subseteq D(C(x))$ *and for each* $d \in D(x)$ *the predecessor of* $C(x)$ *along* d *is the predecessor of* x *along* d *translated by* $(d-1)L^2/(Gd^2)$*: with* $\mathrm{pred}_d(t) = t(d-1)/d^2$*,*

$$\mathrm{pred}_d(C(x)) = \mathrm{pred}_d(x) + \frac{(d-1)L^2}{G\,d^2}.$$

Hence the incoming steps of x *inject into those of* $C(x)$ *(*$d \mapsto d$*) and the collective map acts on the ancestry fan of an integer as a monotone enlargement with explicit translation.*

(d) L3 and L4: restricted positive results, no general map. *On the rigid domain $x = t\sigma^2$, $t \in \{1, 2, 3\}$, σ even, the collective segment $x \to C^3(x) = 4x$ and the edge $T_1 : x \mapsto 4x$ have the same reduced event record, and at integer endpoints the complexes agree (Paper III, Theorem 13.8); for the tower pairs with a SELECT-II ratio the event counts agree in the cases listed in Paper III, Theorem 13.12, and fail for $N = 8$ ($G = 11$) and $N = 3$ ($G = 5$) although the ratios agree. Beyond these cases no map between event records, and none between the history complexes of t and the evolving complexes $K(D(x_n))$, is established; the endpoint geometry cannot be a history invariant (Proposition 7.1), and general constructions are open (O3).*

Proof. (a) and (b) are Theorem 11.2. (c) By Lemma 2.3 $D(x) \subseteq D(C(x))$; by Proposition 4.1(a) the incoming steps of an integer t are indexed by $D(t)$ with predecessor $t(d-1)/d^2$. Therefore each $d \in D(x)$ indexes an incoming step of $C(x)$, and $\mathrm{pred}_d(C(x)) = (x + L^2/G)(d-1)/d^2$, which is $\mathrm{pred}_d(x) + (d-1)L^2/(Gd^2)$. For $d \in D(x)$ the quantity $\mathrm{pred}_d(C(x))$ is an integer because $d^2 \mid C(x)$. (d) The positive statements are Paper III, Theorem 13.8 and Paper III, Theorem 13.12, quoted; the negative statement is Proposition 7.1, and the remaining open questions are in Section 16. (a) is an obstruction to one specific family of maps only: it uses the ratio bound, which a non-identity Φ is not required to respect. □

Remark 11.4. Proposition 11.3(c) is genuinely about the overlap of the two domains, the integers; it uses the generative interpretation of $D(t)$ and the collective step in one formula, but it does not say that $C(x)$ is a generative successor of x, which (a) excludes. We recomputed (c) on 1008 integer states with integer image (Appendix F).

Figure 19 shows the five levels of cross-system correspondence.

12 The unified diagram

12.1 Objects, maps and statuses

The diagram of Figure 21 has two columns of objects. On the generative side: histories, swap classes, contents, endpoints, occurrence sets of integers, endpoint complexes and their Betti numbers, and the route complexes and counts of the history space of a fixed source and endpoint. On the collective side: rational states, their occurrence sets, their move complexes and primitive systems, and the same data at the image under C. Integers belong to both sides.

Every arrow carries one of five statuses (see Section 1 for the line conventions): *established arrow* (a source-defined function, cited or defined here); *proved factorization* (a map that factors through another under stated hypotheses); *proved obstruction* (a reconstruction ruled out by an exact result); *open construction*; *interpretive comparison* (no formal correspondence claimed).

12.2 Factorizations

Figure 20 shows the factorization chain of Sections 3 and 4 with the sets on which it is defined: the maps $\mathrm{Hist}(m_0) \to \mathrm{Cls} \to \mathrm{Cont} \to \mathbb{Z} \to \{D\}$ are for a fixed source m_0 (or all integers in the last step), and $D \mapsto K(D) \mapsto \beta$ are functions of D (Paper III, Theorem 3.4). The two obstructions leaving this chain are the primitive system (Proposition 4.1(c)) and the history counts (Proposition 7.1).

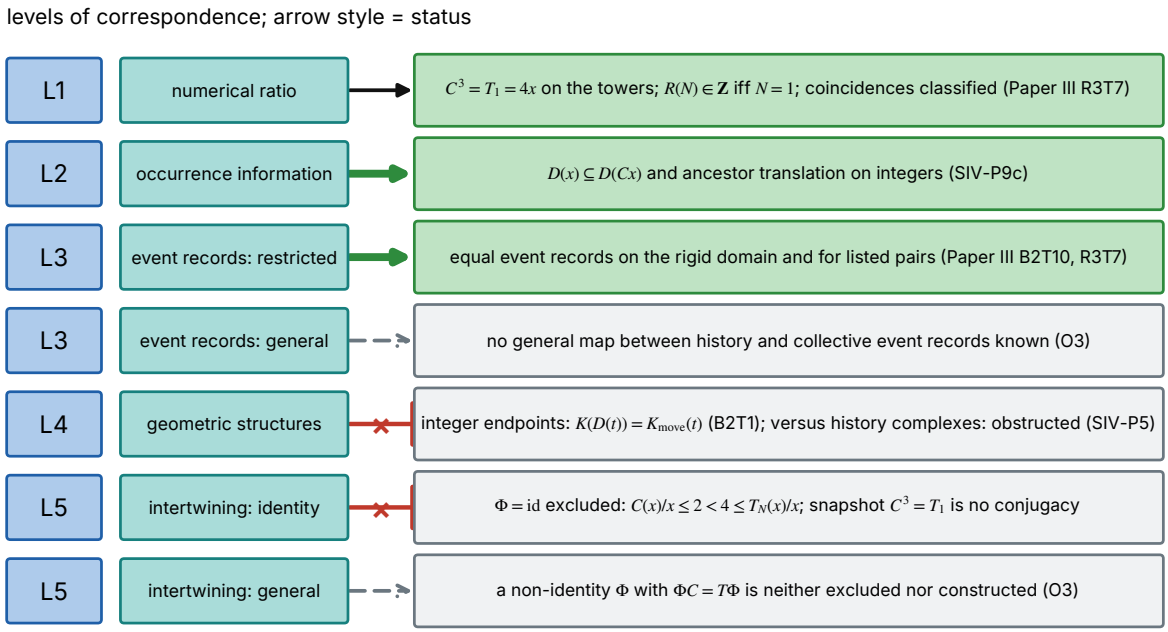


Figure 19: **The five levels of cross-system correspondence.** Arrow style gives the status of what is known: established (L1), proved restricted statements (L2; L3 on the rigid domain and listed pairs), open (L3 in general; L5 for a non-identity Φ), obstructed (L4 against history complexes; L5 for $\Phi = \mathrm{id}$). *Schematic of status; each row cites the proposition or theorem that supports it.*

12.3 Commuting relations

Proposition 12.1 (no claimed commuting squares)**.** *Let $\iota : \mathbb{Z} \to \mathbb{Q}_{>0}$ be the inclusion. The following squares are* not *claimed to commute and the corresponding statements are false or unestablished: (i) $C \circ \iota = \iota \circ T_N$ for any N (false by Proposition 11.3(a); this excludes the identity-type correspondence only, not a non-identity Φ); (ii) $K(D(\cdot)) \circ \mathrm{end}$ determines the history complex (false by Proposition 7.1); (iii) $D \mapsto \mathrm{Gr}$ (does not exist, Proposition 4.1(c)). The only commuting relations proved in this manuscript are the factorization identities* $\mathrm{end} = \pi_3\pi_2\pi_1$ *and* (realised record) $= g \circ D$.

Proof. (i) is Theorem 11.2(i). (ii) is Proposition 7.1. (iii) is Proposition 4.1(c). The two commuting identities are Proposition 3.4(a) and Paper III, Theorem 3.4. □

13 Synthesis propositions and obstructions

Table 1 lists the nine synthesis propositions with their status and what each rests on; the machine-readable register is `crosswalk/SYNTHESIS_PROPOSITIONS.tsv`. A proposition is marked *new* when its statement is derived here with an independent proof; no result is silently upgraded from a finite-exact status.

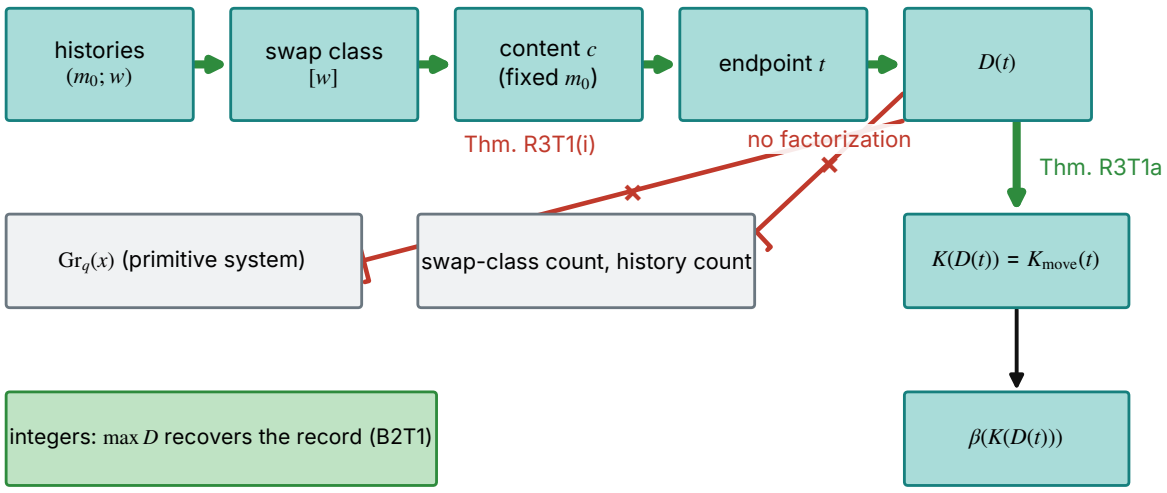


Figure 20: **A source-supported factorization diagram.** What passes through the occurrence set D and what does not. Green: proved factorizations; red: proved obstructions (Paper III, Theorem 3.6(i) and Proposition 7.1). *Exact content: statuses and domains; boxes are schematic.*

13.1 Answers to S1–S4

S1 *Proved for integer states, open beyond.* The generative guard $(N+1)^2 \mid t$ and the occurrence guard $d^2 \mid t$ coincide under $d = N+1$ (Proposition 4.1(a)). For rational states the guard $d \equiv 1 \pmod{b}$ has no generative counterpart; a rational extension of the generative rule reproducing it is open (O8).

S2 *Proved, with obstructions.* The endpoint retains the valuation displacement $V_A c$ and nothing finer (Proposition 3.4); the occurrence set retains everything that enters K_{move} and nothing about the primitive system or about the number of histories (Propositions 4.1 and 7.1). Reconstruction of a history needs the fibre sequence along the path (Theorem 3.2).

S3 *Proved obstruction and open maps.* $K(D(t))$ does not determine the topology of $\mathrm{Hist}(m_0, t)$ (Proposition 7.1); the route-complex side has independent arithmetic (Proposition 6.3); explicit maps between history geometry and evolving occurrence geometry beyond L2 are open (Proposition 11.3, O3).

S4 *Conditional (structural).* Both criteria are instances of monotone minimality (Corollary 5.5); an exact implication between them requires an identification of ambient orders and is open (O7).

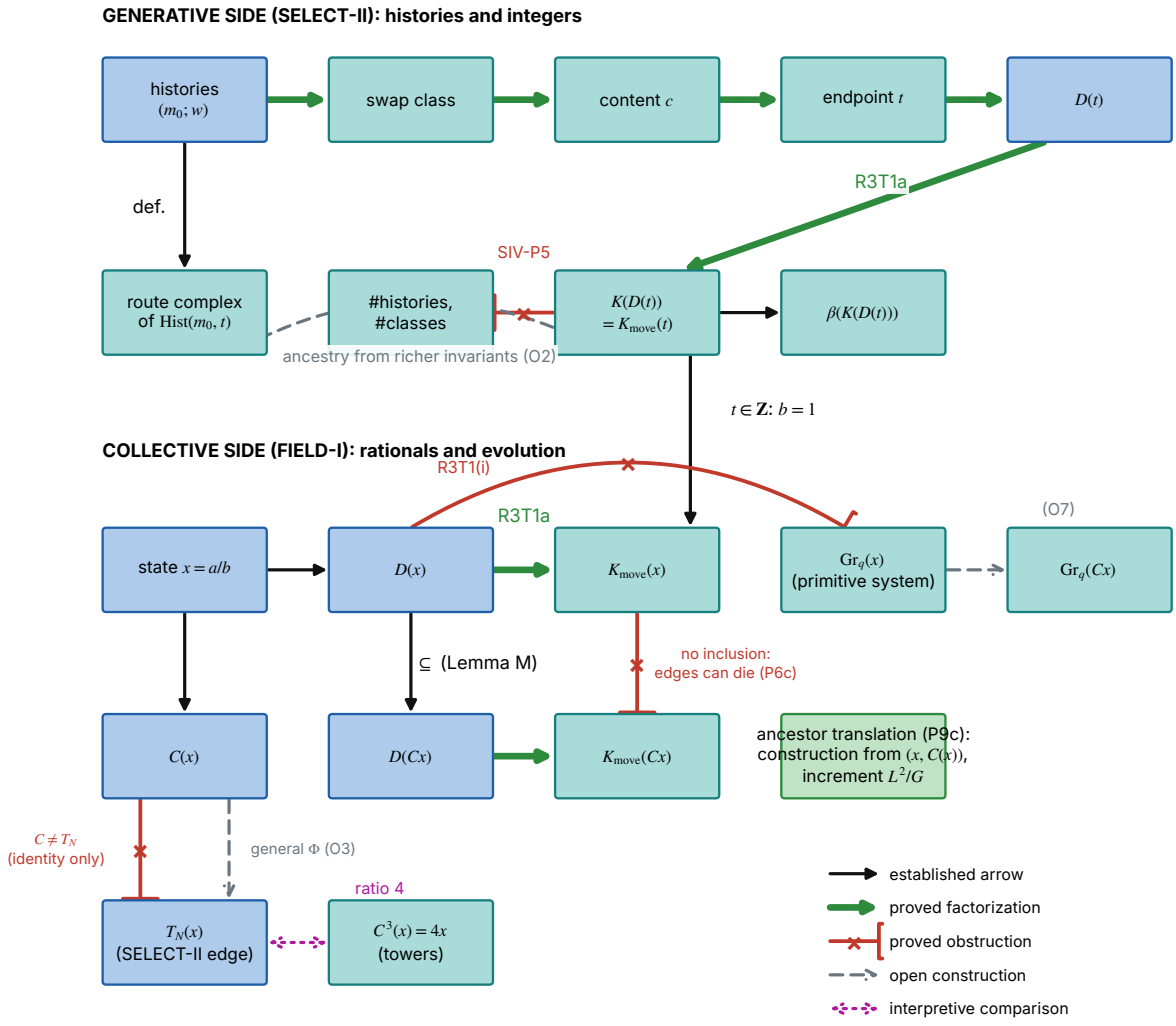


Figure 21: **The global diagram of maps, factorizations and obstructions.** Objects are the structures of Section 2; each arrow is labelled with its status by line style and cited result. Green arrows: factorizations proved from the cited theorems; black arrows: established functions; red arrows with a cross: proved obstructions (R3T1(i) is Paper III, Theorem 3.6(i); SIV-P5 is Proposition 7.1; $C \neq T_N$ is Proposition 11.3(a)); grey dashed arrows: open constructions (O2, O3, O7); purple dotted arrow: interpretive comparison of the ratio 4 (Theorem 11.2(ii)). The arrow between $K_{\mathrm{move}}(x)$ and $K_{\mathrm{move}}(Cx)$ is drawn as an obstruction to an *inclusion*: vertices persist but edges can be lost (Proposition 8.2(c)), so no complex inclusion is implied. The box *ancestor translation* is a construction from the arithmetic pair $(x, C(x))$ and the set $D(x)$ (Proposition 11.3(c)), not a function of $K_{\mathrm{move}}(Cx)$, and is therefore drawn without an arrow from that complex. The legend is in the figure. *Schematic of status: the diagram is a map of established relations, not a commuting diagram; the status of each arrow is defined in Appendix D.*

ID	Subject	Status	Statement
SIV-P1	Endpoint projection and retained information	NEW-PROVED	Proposition 3.4
SIV-P2	Occurrence projection and primitive-relation information	NEW-PROVED (a); CITED (b),(c)	Proposition 4.1
SIV-P3	Arithmetic availability versus geometric realization	NEW-PROVED (a),(b); CITED (c),(d)	Proposition 5.3
SIV-P4	Arithmetic constraints and global history connectivity	NEW-PROVED from cited theorems; FINITE check	Proposition 6.3
SIV-P5	Limits of endpoint geometry as a history invariant	NEW-PROVED (exhaustive recursion); FINITE-EXACT census	Proposition 7.1
SIV-P6	Hierarchy of structural persistence under evolution	NEW-PROVED; (d) quotes finite-exact census	Proposition 8.2
SIV-P7	Information requirements for topological prediction	NEW-PROVED (a)-(d); (e) search outcome only	Proposition 9.1
SIV-P8	Scaling information versus geometric structure	NEW; FINITE-EXACT (k¡=14, 12 periods)	Proposition 10.1
SIV-P9	Correspondence levels for distinct arithmetic dynamics	NEW-PROVED (c); ASSEMBLED (a),(b),(d)	Proposition 11.3

Table 1: **Register of the synthesis propositions.** New results are proved in the text or in Appendix C; inherited ingredients are cited by theorem number (Appendix B); finite-exact statements are labelled with their checked ranges.

14 Relation to established theories

Each comparison is classified as **identical** (an established equivalence), **special case** (a proved inclusion), **analogy** (similar behaviour without a proved equivalence) or **open**. Nothing in the manuscript is proved by appeal to these theories, and no claim of priority is made.

Arithmetic dynamics and divisibility systems. The map C is a rational map of the positive rationals defined by arithmetic data; its occurrence set is a divisor condition on the numerator. *Analogy*: we do not connect C with the standard theory of dynamics of rational functions; the comparison with p-adic or arithmetic-dynamical invariants is *open*.

Geometry of numbers and lattice relations. The lattice $\Lambda_b \subseteq \mathbb{Z}^k$ of exponent vectors with product $\equiv 1 \bmod b$ is the kernel of a homomorphism to $(\mathbb{Z}/b)^\times$; its transport (Paper I, Theorem 3.2) is a statement about kernels of homomorphisms. *Identical* with the standard description of such sublattices (cf. Cassels, *An Introduction to the Geometry of Numbers*); the dimensional results are UNNS-specific.

Graver bases. The set Gr_q is the Graver basis of the lattice restricted to a box [9, 10], and the persistence criterion Paper I, Theorem 4.2 is a statement about conformal minimality. *Special case* of the notion of a Graver basis; the monotone-minimality principle (Lemma 5.2) is a general, elementary statement which these criteria instantiate.

Rewriting systems and confluence; trace monoids. Swap classes are the classes of a rewriting system generated by commutation of adjacent letters when both orders are admissible, as in partially commutative (trace) monoids [12]. The commutation here is *state-dependent*, so a history space is not a free partially commutative monoid; *analogy*.

Confluence of the block rewrites is studied in Paper II; whether the swap graph can be described by a trace-type normal form is *open*.

Cubical and cellular complexes; homology. The complexes $K(D)$ are cubical in the sense of Paper III's definition; homology is computed over $\mathbb{F}_2$ (and checked over another field in the sources), and cellular homology and the long exact sequence of a pair are standard [13]. *Identical* use of standard tools; the divisor-interval cell structure is specific.

Zig-zag and filtered structures. The shape $K \leftarrow K^\circ \rightarrow K'$ of an event resembles a zig-zag diagram of complexes [14]; we use only that shape. *Analogy*; no use is made of zig-zag persistence theory.

Information-losing projections. The factorization lemma Lemma 2.5 is the universal property of a quotient by the fibre relation of a surjection; *identical*.

15 Principal mathematical conclusions

Each conclusion cites the result on which it rests; broad readings are labelled as interpretation.

15.1 Operational characterisation of the UNNS Substrate

The preceding results support an operational characterisation of the UNNS Substrate as a rule-governed generative arithmetic architecture. Its mathematical organisation is not exhausted by numerical states, individual generating histories, occurrence inventories, primitive arithmetic relations, or realised geometric complexes considered separately. It is expressed through the admissible transformations and structural relationships among these objects.

The generative framework may be represented schematically by

$$\mathcal{U} = (S, \mathcal{O}, \mathrm{Dom}, \mathcal{R}, \mathsf{Adm}),$$

where S is a set of states, $\mathcal{O}$ a set of operators, Dom the assignment of a domain to each operator, $\mathcal{R}$ the set of admissible routes, and Adm the conditions governing admissibility. In the arithmetic constructions studied here, S consists of the positive integers and rationals, $\mathcal{O}$ of the maps T_N and C, Dom of the conditions $N \mid m$ and $D(x) \neq \emptyset$, $\mathcal{R}$ of histories and collective orbits, and Adm of the divisibility conditions on squares. The tuple is a notation for organising what has been proved; it is not an axiom system from which the results are derived. The occurrence and geometric structures studied here (D, the lattice of relations, the primitive system Gr_q, the complexes $K(D)$ and their Betti numbers) are associated constructions, not additional primitive components assumed without definition.

The results establish the following characteristic properties of this architecture; each is a restatement of the cited result for the stated domain.

Property	Support
Numerical endpoints need not identify their generative histories; the endpoint retains only the valuation displacement.	Proposition 3.4, Proposition 7.1
Occurrence inventories determine the realised move complexes, but not the complete primitive relation systems.	Proposition 4.1(b),(c); Paper III, Theorem 3.4, Paper III, Theorem 3.6
Arithmetic relations can remain dormant until further occurrences permit their geometric realisation.	Proposition 5.3; Paper I, Theorem 4.2
Global connectivity of the history space is governed by arithmetic of the source, not by local swaps alone.	Proposition 6.3
Under the collective map, occurrence sets grow, connectivity only coarsens, and each edge exists during an arithmetically determined interval.	Proposition 5.3(b), Proposition 8.2(a)–(c)
Event steps follow exact arithmetic laws, and at graph level the change of the first two Betti numbers follows from local counts and prime supports; local counts alone do not determine it, and beyond graph level the determination is open.	Proposition 9.1; Paper III, Theorem 7.9
Scaling parameters do not determine topology (finite-exact).	Proposition 10.1
The two rules are related at integers and occurrence labels, not by a one-step identity correspondence.	Proposition 11.3; Figure 21

These properties show that generation, arithmetic possibility, geometric realisation and structural evolution are distinct but mathematically related aspects of the framework. Their relationships are governed by the proved factorizations, admissibility conditions, transition laws and obstructions of Sections 3 to 11, collected in Figure 21.

Consequently, the UNNS Substrate is characterised operationally not by a single geometric object, but by a system of admissible generative mechanisms and the layered structures they produce and transform.

Scope. This characterisation is supported by the results established for the specified arithmetic constructions. It does not assert that every component derives from a unique minimal axiom system, that all generative operators are dynamically equivalent, or that the mathematical framework has been identified with a physical substrate. Establishing a fully intrinsic characterisation remains a separate mathematical problem (Section 16.1).

15.2 Mathematical implications

Each conclusion cites the result on which it rests; broad readings are labelled as interpretation.

C1. Generation. Arithmetic conditions decide which steps and which histories exist: a step $m \xrightarrow{N} t$ exists iff $d = N+1$ satisfies $d^2 \mid t$ (Proposition 4.1(a)), and legal swaps are decided by divisibility at each state (Proposition 6.3).

C2. Ancestry. The numerical endpoint determines only the valuation displacement of the history (Proposition 3.4); equal occurrence sets are compatible with different numbers of histories and of swap classes (Proposition 7.1).

C3. Geometry. The realised geometry of a state is a function of its occurrence set (Paper III, Theorem 3.4); the primitive relation system contains information that the occurrence set does not (Proposition 4.1(c)).

C4. Transport. Preservation under lattice inclusion holds in rank two (Paper I, Theorem 7.5) and fails in rank three under the hypotheses of Paper I, Theorem 8.2; abstract and realised loss differ (Paper I, Proposition 6.3).

C5. Recombination. Local legal swaps do not force a connected history space: the class count of $\{2, 3, 14\}$ is 0, 1 or 2 according to arithmetic of the source and is not monotone in divisibility (Proposition 6.3).

C6. Evolution. Collective steps enlarge the occurrence set, only coarsen the connectivity relation, give each edge an interval of existence, and leave higher Betti numbers unconstrained by monotonicity (Propositions 8.2 and 5.3).

C7. Prediction. Arithmetic congruences give the event step; supports and local counts give $\Delta\beta_0$ and, at graph level, $\Delta\beta_1$; positive-degree homology needs attachment data (Proposition 9.1, Paper III, Theorem 7.9, Paper III, Theorem 10.1). Scaling data do not determine topology (Proposition 10.1).

C8. Compatibility. The one-step identity correspondence of the two rules is obstructed (a non-identity intertwining is open); the overlap on integers admits a translation formula; the other levels are obstructed or open (Proposition 11.3).

Answer to the governing question. *Interpretation, supported by C1–C8 and not a theorem:* in the part of the theory that is established, geometry is formed from arithmetic by two successive forgetful maps—from histories to endpoints, from states to occurrence sets—and the formation of geometry is stable in a precise sense (occurrence sets only grow; connectivity only coarsens) but not rigid (edges and higher homology are created and destroyed). Where the two generating rules meet, they meet in integers and in occurrence labels, and the exact obstructions show that nothing richer follows from the labels alone.

15.3 Limitations

The conclusions concern the specified arithmetic constructions and the finite-exact ranges stated in the propositions. Verification inherited from Papers I–III is not re-run here (NOT_RERUN). The collective rule is studied on its recorded domains and the generative rule on integer sources; the extension of the generative rule to rational sources is open (O8). No claim is made that the operators T_N and C belong to one dynamical system, that the associated constructions are independent of one another beyond the stated obstructions, or that the characterisation applies outside the constructions named. Constant C3 is unclaimed.

16 Open questions

Each problem states the nearest proved theorem and the missing obligation. Problems already settled by the cited releases are not repeated.

O1. Classification of history complexes. *Nearest:* the fixed-content classification of $\{2, 3, 14\}$ (Paper II, Theorem 5.8) and the scoped block rewrites (Paper II, Theorem 10.2). *Missing:* a classification of swap classes and route homology for arbitrary content and source.

O2. Reconstruction of ancestry from richer invariants. *Nearest:* faithfulness of the fibre sequence (Theorem 3.2) and the obstruction Proposition 7.1. *Missing:* a characterisation of the invariants of an endpoint, strictly between $D(t)$ and the whole fibre sequence, that determine the number of histories or of classes.

O3. Maps between the two geometries. *Question:* maps between history geometry and evolving occurrence geometry, and non-identity intertwining of the two dynamics. *Nearest:* the translation formula Proposition 11.3(c) and the restricted event correspondences Paper III, Theorem 13.8, Paper III, Theorem 13.12. *Missing:* a map at level L3 or L4 beyond the rigid domain and the listed pairs, and either a construction or a non-existence proof for a non-identity Φ with $\Phi \circ C = T \circ \Phi$ (the obstruction Proposition 11.3(a) excludes only $\Phi = \mathrm{id}$).

O4. Prediction of higher-dimensional Betti changes. *Nearest:* Proposition 9.1(c) and the zig-zag identity. *Missing:* determination of $\Delta\beta_i$, $i \geq 1$, when 2-cells are present, from data cheaper than the full target.

O5. Effective onset and closed Betti laws for towers. *Nearest:* Paper III, Theorem 12.4 (qualitative quasi-polynomials) and Paper III, Theorem 12.10 (exact to $k \leq 140$), Proposition 10.1. *Missing:* an effective threshold, exact formulas for all k, and an invariant of S_0 that governs the Betti vector.

O6. Reachability of activation families. *Nearest:* Paper III, Theorem 4.3 and Paper III, Theorem 8.5. *Missing:* a complete arithmetic classification of the families that occur along actual orbits.

O7. Transition laws for dormant primitive systems. *Nearest:* Paper I, Theorem 4.2, Lemma 5.2, Corollary 5.5. *Missing:* an identification of the order on lattice vectors with the divisor-interval order on pairs of occurrences under compatible hypotheses, or an obstruction to one.

O8. Compatibility of arithmetic guards. *Nearest:* the bijection Proposition 4.1(a) at the level of integers. *Missing:* an extension of the generative rule to rational sources whose predecessor guard reproduces $d \equiv 1 \pmod{b}$.

Figure 22 shows established conclusions and open construction obligations.

16.1 Operational versus intrinsic characterisation

Section 15.1 characterises the Substrate *operationally*: through the operators, domains, routes and admissibility conditions that are actually used, and the proved relations among the structures they generate. An *intrinsic* characterisation would instead derive these components, and the associated constructions, from a minimal list of axioms and show which of them are independent. The two are different mathematical problems, and only the first is addressed here. *Nearest proved results:* the factorizations and obstructions of Propositions 4.1, 7.1 and 3.4 and the status map of Figure 21. *Missing obligations:*

(i) an axiomatic definition of admissibility from which the guards $N \mid m$ and $d^2 \mid a$, $d \equiv 1$ (b) both follow (the identification is proved only for integer states, Proposition 4.1(a); see O8);

each open item is stated in Section 16 with its nearest proved theorem and missing obligation

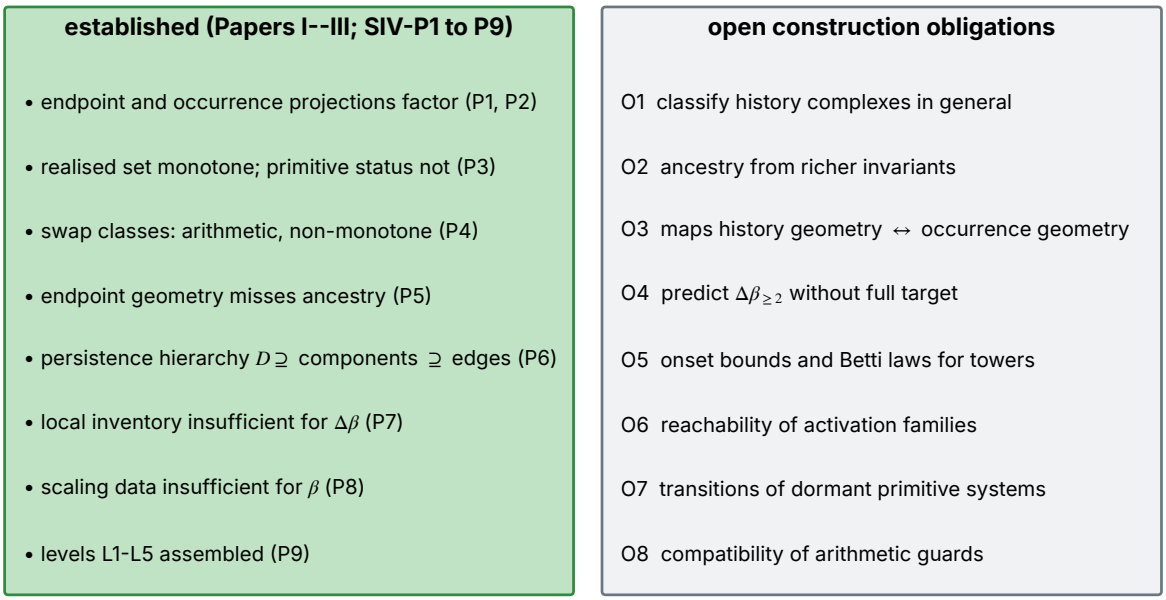


Figure 22: **Established conclusions and open construction obligations.** Left: the nine propositions; right: the eight open problems of this section. *Schematic summary; each item is defined and referenced in the text.*

(ii) a compatibility statement placing T_N and C in one admissibility system, or a proof that no non-identity intertwining exists (O3);

(iii) a proof that the associated constructions (occurrence set, relation lattice, primitive system, complex) are determined by, or independent of, the schematic components $(S, \mathcal{O}, \mathrm{Dom}, \mathcal{R}, \mathsf{Adm})$, rather than assumed through definitions;

(iv) the passage from the finite-exact ranges recorded here to statements for all sources, which no result of this paper provides.

References

[1] UNNS Research Collective, *Arithmetic Transport and Dimensional Obstructions in UNNS Occurrence Geometry*, Manuscript I of the programme GEO-HD-1, version 0.4 (2 Nov. 2026); file `PAPER_I_TRANSPORT_v0.4.pdf`, SHA-256 `19fc10f546696909eef39c75ebdb47b5468d93173f9bd9e82567bd5f60f9eef9`.

[2] UNNS Research Collective, *Generative Histories, Arithmetic Recombination, and Higher-Dimensional Route Topology*, Manuscript II of the programme GEO-HD-1, version 0.4 (2 Nov. 2026); file `PAPER_II_HISTORIES_v0.4.pdf`, SHA-256 `df34fa94fed9fb33b82c87726ca9e725307723f4e3cb67e1af2f55d7b56801e4`.

[3] UNNS Research Collective, *Geometric Evolution under Collective UNNS Dynamics: Activation, Events, and Topological Prediction*, Manuscript III of the programme GEO-HD-1, version 0.4 (3 Nov. 2026); file `PAPER_III_EVOLUTION_v0.4.pdf`, SHA-256 `47fae77fd9c11f0cc7e81447fd697cdbae413e2646564b1d966836ebad14cb8e`.

[4] UNNS Substrate programme, *Mathematical Foundations of the UNNS Substrate, Volume I: Generative Arithmetic and Structural Reachability* (the grammar SELECT-II and the occurrence set). Source: `ukbbi/UNNS`, commit `9a021346b2b3d964523440d2737b9d0c4ddbb3e5`, `projects/UNNS_MATH_FOUNDATIONS/Manuscript`.

[5] UNNS Substrate programme, *Mathematical Foundations of the UNNS Substrate, Volume II: Collective Dynamics, Prime Rigidity, and Effective Predictability* (the collective map FIELD-I). Same commit and path as [4].

[6] GEO-I, Papers A and B of the geometry programme (occurrence geometry, region-Graver sets and the realised complexes), cited at commit `9a021346b2b3d964523440d2737b9d0c4ddbb3e5` as in [4].

[7] Foundations reference collection v001 (SELECT-I/II, FIELD-I v001–v012; frozen), `ukbbi/UNNS`, commit `bb8b12d753c600c597feb00df17fe139c7cc65fa` (23 Oct. 2026), `projects/UNNS_MATH_FOUNDATIONS/sources`.

[8] GEO-HD-1 research collection, snapshot `a88f31487f3b05c9db297dcd1192d56eea2b0c7d` (the pin named in the programme specification; not independently verified in this work). Frozen releases from which the inherited records are taken, with SHA-256 as computed on the files supplied for Paper III:

- `GEO_HD_1D_v001` `c532b7410f3bf9d437f9f172b3e56029cd9f7bab5b2ac0c0e533713155eaf7a4`; `GEO_HD_1D_v002` `8baf463c07bb5285dd53b355aee05a0f841c068ba3ae4794f3702adff6a4bae0`; `GEO_HD_1D_v003` `65dbb281e4c75d625b6cf3e4bcb8262d7d42e68d0c476fbba225fcde744a4706`; `GEO_HD_1D_v004` `e0390e2a2f88eaeb11042de09fd056741ed98f94318de708d308faa090f05b5b`;
- `GEO_HD_BRIDGE_v001` `e99472809928f4a23fb292f60759965159c7a4db5f018d56748ccac328719395`; `v002` `012ead97e5b8854cd5218417d46f0e388d927b6d59d5f6e75a8cc343f5dd4955` (with errata for B2-T9 and B2-T7); `v003` `6fba87af9f3096772e43f4ac77fa8d55aea9e242eeef241081a987b567d5d5a8`; `v004` `cfb31157a39666d1a4bbb33b280ec626c7de2c141e8232da43c8b934b8affff4`; `v005` `50e342821d7a5f760c5127df348e97f3a4d4d3bf34bf60e1019d6336f28c5d03`.

[9] J. E. Graver, On the foundations of linear and integer linear programming I, *Math. Programming* **9** (1975), 207–226.

[10] B. Sturmfels, *Gröbner Bases and Convex Polytopes*, University Lecture Series 8, American Mathematical Society, 1996 (Graver bases, §4).

[11] J. W. S. Cassels, *An Introduction to the Geometry of Numbers*, Grundlehren der mathematischen Wissenschaften 99, Springer, 1959 (lattices; sublattices as kernels).

[12] V. Diekert, G. Rozenberg (eds.), *The Book of Traces*, World Scientific, 1995 (partially commutative monoids).

[13] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002 (cellular homology, §2.2; long exact sequence of a pair, §2.1).

[14] G. Carlsson, V. de Silva, Zigzag persistence, *Found. Comput. Math.* **10** (2010), 367–405 (zig-zag diagrams of complexes; cited for the shape $K \leftarrow K^{\circ} \rightarrow K'$ only).

On the references. Papers I–III are cited at version 0.4 by theorem number; every such number is generated from the corresponding `.aux` file (`scripts/gen_xref.py`), so a citation that does not resolve is reported at build time. UNNS foundation sources are cited at the commits and archive hashes fixed by the programme; repository access during this work was read-only and the pinned repositories were not fetched. External literature is cited only for the standard notions named and for resemblance of structures; no statement is proved by appeal to an equivalence with those theories.

A Notation

Symbol	Meaning (source)
$T_N(m)$, $R(N)$	$m(N+1)^2/N$ $(N \mid m)$; $R(N) = (N+1)^2/N$ (Foundations I)
$\mathrm{Hist}(m_0)$, $\mathrm{Hist}(m_0, t)$	histories from m_0; those ending at t (Paper II)
$c(h)$, $F(h)$, $\mathrm{end}(h)$	content, fibre sequence, endpoint of a history
V_A, $\Delta_p(N)$	displacement matrix, $\Delta_p(N) = 2v_p(N+1) - v_p(N)$ (Paper II)
$\nu(m)$	number of swap classes of $\{2, 3, 14\}$ at source m
$D(x)$	$\{d \geq 2 : d^2 \mid a,\ d \equiv 1\ (b)\}$ for $x = a/b$ (Foundations II)
L, G, $C(x)$	$\operatorname{lcm} D$, $\gcd\{d-1\}$, $x + L^2/G$
$I(u, v)$	$\{z : \gcd(u, v) \mid z \mid \operatorname{lcm}(u, v)\}$ (divisor interval)
$K(D)$, $K_{\mathrm{move}}(x)$	divisor-interval complex; $K_{\mathrm{move}}(x) = K(D(x))$ (Paper III)
Λ_b, $\mathrm{Gr}_q(x)$	relation lattice; region Graver set (Paper I)
$\mathrm{cl}(D)$	number of classes of the share-a-prime relation on D
π_1, π_2, π_3	projections of Section 3
$\beta = (\beta_0, \beta_1, \dots)$	Betti numbers over $\mathbb{F}_2$
$a(u)$, $p(u, v)$	arrival time of u; first piercing time of $I(u, v)$ (Proposition 5.3)

B Theorem provenance

The table below lists every result of Papers I–III used in this manuscript with its number in the v0.4 manuscript (generated from the `.aux` files), the crosswalk identifier (UID) and source identifier recorded there, the source status, and the role the result plays here: *cited* (used as stated), *used to derive* (an ingredient of a new proof) or *proved again* (a self-contained proof is given because the statement is needed in a modified form). Crosswalk UIDs are copied unchanged. The machine-readable form is `crosswalk/PAPER_IV_SOURCE_MAP.tsv`.

Errata. The accepted errata of the frozen releases (for B2-T7 and B2-T9) concern statements that this manuscript does not use; Paper III's treatment of the corrected B2-T9 is not relied upon. The results used here are cited with the hypotheses recorded in the table; finite-exact results are not promoted to theorems.

Source	UID	ID	Status	Role in Paper IV	Hypotheses / corrections
Paper II Definition 2.1	none	-	-	definition (cited) (Sec.2)	source Foundations Vol. I
Paper II Definition 2.4	CW-0337	D-W	PROVED	definition (cited) (Sec.2)	
Paper II Theorem 3.1	CW-0336	D-T0	PROVED	proved again (self-contained proof in SIV-P2(a)) (SIV-P2,SIV-P5,SIV-P9)	integer t
Paper II Theorem 9.1	CW-0369	D4-T1	PROVED	cited as ingredient (SIV-P1)	fixed source, finite alphabet
Paper II Theorem 11.3	CW-0341	D-T4	PROVED	cited (SIV-P5 (d), S2)	fixed source
Paper II Theorem 5.8	CW-0348	D-T10	PROVED	used to derive (SIV-P1, SIV-P4)	content 2,3,14
Paper II Theorem 11.9	CW-0413	R3-T3	PROVED-SCOPED	cited (SIV-P4 context)	source 4, A disjoint from 1,2,4; proved-scoped; scope as in Paper II
Paper II Theorem 8.2	CW-0340	D-T3	PROVED	used to derive (SIV-P4(d))	r=2 used
Paper II Theorem 10.2	CW-0371	D4-T3	PROVED	cited (O1)	scoped block rewrites
Paper II Theorem 7.5	CW-0361	T-SPHERE	PROVED	cited (Sec.6)	scoped
Paper II Theorem 7.3	CW-0359	T-DUAL	PROVED	cited (Sec.6)	
Paper III Definition 2.1	none	-	-	definition (cited) (Sec.2)	source Foundations Vol. II
Paper III Lemma 2.6	CW-0425	A4-T3a	PROVED	used to derive (SIV-P3,P6,P9)	all reduced rationals with D nonempty
Paper III Definition 9.2	none	-	-	definition (cited) (Sec.2)	
Paper III Theorem 3.4	CW-0409	R3-T1a	PROVED	used to derive (SIV-P2(b), Fig.20)	reduced rational states
Paper III Theorem 3.6	CW-0410	R3-T1	PROVED	cited (obstruction) (SIV-P2(c))	r¿=13 prime, r=5 mod 8
Paper III Theorem 3.1	CW-0395	B2-T1	PROVED	cited (SIV-P2(b))	integer m with sigma¿1
Paper III Theorem 9.1	CW-0441	A5-T3.1	PROVED	used to derive (SIV-P3,P6; Lemma comp)	any finite D
Paper III Theorem 4.3	CW-0423	A4-T1	PROVED	cited (example) (SIV-P3(c))	k¿=2, l prime with k—l-1, ...
Paper III Theorem 8.5	CW-0440	A5-T2.4	PROVED	cited (Sec.9)	activation family before first event
Paper III Theorem 7.9	CW-0437	A5-T2.1	PROVED	cited (Sec.9, Fig.13)	stage Y_j
Paper III Corollary 7.10	CW-0438	A5-T2.2	PROVED	cited (Sec.9)	
Paper III Theorem 10.1	CW-0444	A5-T4.1	PROVED	cited (SIV-P7 remark, Sec.9)	any event K ¡- K0 -¿ K'
Paper III Theorem 10.6	CW-0446	A5-T4.3	PROVED	cited (Sec.9)	Dead empty
Paper III Theorem 10.4	CW-0445	A5-T4.2	PROVED	cited (SIV-P6)	
Paper III Theorem 11.6	CW-0452	A5-T7	MIXED	cited (finite-exact census) (SIV-P6(d))	l in 5,7,11,13, S¡1500, 24 steps; mixed record; census only
Paper III Theorem 6.2	CW-0419	R3-T5c	PROVED	cited (SIV-P8, Sec.10)	l=5 tower
Paper III Theorem 6.1	CW-0418	R3-T5b'	PROVED	cited (Sec.10)	l=11 tower
Paper III Theorem 12.4	CW-0450	A5-T5.3	PROVED-SCOPED	cited (Sec.10, O5)	qualitative; no effective threshold; proved-scoped

Source	UID	ID	Status	Role in Paper IV	Hypotheses / corrections
Paper III Theorem 12.10	CW-0451	A5-T6	FINITE-EXACT	cited (finite-exact) (SIV-P8)	k¡=140; finite-exact; not extrapolated
Paper III Proposition 13.6	none	-	-	cited (derived in Paper III; no separate cross-walk row) (SIV-P9(a))	D(x) nonempty
Paper III Theorem 13.12	CW-0422	R3-T7	PROVED	cited (SIV-P9(b))	towers
Paper III Theorem 13.3	CW-0397	B2-T3	PROVED	cited (Remark after SIV-P1)	
Paper I Theorem 3.2	CW-0467	P2-T1	PROVED	cited (Sec.2)	
Paper I Theorem 4.2	CW-0469	P2-T2	PROVED	used to derive (instance of minimality lemma) (SIV-P3, Cor. S4)	
Paper I Theorem 4.3	CW-0470	P2-T3	PROVED	cited (Sec.2)	
Paper I Theorem 6.4	CW-0465; CW-0475	E2-11; E2-P2-V	COUNTEREXAMPLE; PROVED	cited (counterexamples) (SIV-P3(d))	
Paper I Theorem 7.5	CW-0504; CW-0497	P5-P1; P4-P1	PROVED; PROVED	cited (Sec.5, Fig.8)	rank 2, any box
Paper I Theorem 8.2	CW-0508	P5-C1	PROVED	cited (Sec.5, Fig.8)	E3, box (2,4,2)
Paper I Theorem 5.2	CW-0479	P3-T3	PROVED	cited (Sec.5)	
Paper I Theorem 9.1	CW-0511	P5-S4	PROVED	cited (Sec.5)	free coordinate
Paper I Proposition 6.3	CW-0474	P2-G	FINITE-EXACT	cited (finite-exact) (Sec.5, Fig.9)	specimen 21ˆ8/5 -¿ 21ˆ8/4; finite-exact; not recomputed

C Proofs of the synthesis lemmas

Proof of Lemma 8.1 (component rule). Write $u \approx v$ if $\gcd(u,v) > 1$. *Edges are $\approx$-pairs.* If $\{u,v\}$ is an edge and $\gcd(u,v) = 1$ then $1 \in I(u,v) \cap (D \cup \{1\})$ and $1 \neq u, v$ (as $u, v \geq 2$), contradicting the interval rule Theorem 5.4. Hence components of $K(D)$ lie inside $\approx$-classes.

$\approx$-pairs are connected. We show: if $u, v \in D$, $u \neq v$, $\gcd(u,v) = g > 1$, then u, v lie in one component, by induction on $|I(u,v)|$. If $I(u,v) \cap (D \cup \{1\}) = \{u,v\}$ the pair is an edge. Otherwise there is $w \in D \cup \{1\}$, $w \notin \{u,v\}$, $w \in I(u,v)$; $w \neq 1$ because $g \mid w$ and $g > 1$. So $w \in D$, $g \mid w$, $w \mid l := \mathrm{lcm}(u,v)$. Then $\gcd(u,w) =: g'$ is a multiple of g and $\mathrm{lcm}(u,w) =: l'$ divides l, so $I(u,w) \subseteq I(u,v)$. If $g' = g$ and $l' = l$, write $u = ga$, $v = gb$ with $\gcd(a,b) = 1$, so $l = gab$; $w = gc$ with $\gcd(a,c) = 1$ (from $g' = g$) and $\mathrm{lcm}(a,c) = ac = ab$ (from $l' = l$), so $c = b$ and $w = v$, a contradiction. Hence $I(u,w) \subsetneq I(u,v)$ and, since $g' > 1$, the induction hypothesis connects u and w. Symmetrically w and v are connected, hence u and v. Thus components of $K(D)$ are the classes of the equivalence relation generated by $\approx$, which is the second formulation (the bipartite graph between D and the primes dividing its elements has the same components). □

Proof of Proposition 5.3(b), restated. See the proof in Section 5; it uses only monotonicity of D_n and the interval rule. □

Proof of the translation formula, Proposition 11.3(c). Given in Section 11; the only arithmetic input is $D(x) \subseteq D(C(x))$ and the bijection of Proposition 4.1(a). □

Proof of Proposition 9.1(c), restated with the cell counts. For a finite complex $\chi = \sum(-1)^i f_i = \sum(-1)^i \beta_i$. If K and K' have no 2-cells, $\beta_0 - \beta_1 = f_0 - f_1$ for each, so $\Delta\beta_1 = \Delta\beta_0 - \Delta f_0 + \Delta f_1$. □

D Diagram definitions

The table below defines each object and arrow of Figure 21 with its status and justification. "Domain" is the set on which the map is taken. No square of the diagram is claimed to commute (Proposition 12.1).

Arrow	Status	Domain / codomain	Justification
Hist → swap class → content → endpoint	factorization	Hist(m_0) (fixed source) to $\mathbb{Z}$	Proposition 3.4(a)
endpoint → $D(t)$	factorization	$\mathbb{Z}$ → finite subsets	definition of D
$D(t) \to K(D(t)) \to \beta$	established	finite D → complexes	Paper III, Theorem 3.4
$D(t) \not\to$ (#hist, #classes)	obstruction	integers, source 4	Proposition 7.1
Hist → route complex	established (definition)	Hist(m_0, t)	Paper II
$K(D(t)) \dashrightarrow$ route complex	open (O2)	–	no map claimed
$t \in \mathbb{Z}$: $K_{\text{move}}(t) = K(D(t))$	established	integers $\subset \mathbb{Q}_{>0}$	$b = 1$
$x \to D(x) \to K_{\text{move}}(x)$	established, factorization	reduced rationals	Paper III, Theorem 3.4
$D(x) \not\to \mathrm{Gr}_q(x)$	obstruction	rationals with equal D	Paper III, Theorem 3.6(i)
$\mathrm{Gr}_q(x) \dashrightarrow \mathrm{Gr}_q(Cx)$	open (O7)	–	Corollary 5.5
$x \to C(x)$; $D(x) \subseteq D(Cx)$	established	rationals	Paper III, Lemma 2.6
$K_{\text{move}}(x) \to K_{\text{move}}(Cx)$ edge rule	factorization	edges between old vertices	Proposition 5.3(b)
translation of ancestors	established (new)	integers x, $C(x)$ integer	Proposition 11.3(c)
$C \neq T_N$	obstruction	$\{x : D(x) \neq \emptyset\}$	Paper III, Proposition 13.6
$T_N - C^3$ ratio 4	interpretive	$G = 5$ towers	Paper III, Theorem 13.12; no formal correspondence

E Counterexamples and obstructions

Each entry gives the exact witness; all were recomputed (Appendix F).

E1 (endpoint ≠ content). Source 32: $(2, 2, 2, 2)$ and $(1, 8, 8)$ both end at 13122; lengths 4 and 3.

E2 (content ≠ class). Source 14: $(14, 3, 2)$ and $(2, 3, 14)$ both end at 5400, two classes.

E3 (equal D, different ancestry). $D(900) = D(9000) = \{2, 3, 5, 6, 10, 15, 30\}$ and $D(288) = D(4320)$; histories and classes from source 4:

target	histories from source 4	swap classes
900	$(2,9,2)$, $(2,2,9)$	1: $\{(2,2,9), (2,9,2)\}$
9000	$(4,5,9,2)$, $(2,9,5,4)$, $(2,9,4,5)$, $(4,5,2,9)$	2: $\{(4,5,2,9), (4,5,9,2)\}$; $\{(2,9,4,5), (2,9,5,4)\}$
288	$(2,1,1)$, $(1,2,1)$, $(1,1,2)$	1: $\{(1,1,2), (1,2,1), (2,1,1)\}$
4320	$(4,5,3,2)$, $(4,5,2,3)$, $(2,3,4,5)$	2: $\{(4,5,2,3), (4,5,3,2)\}$; $\{(2,3,4,5)\}$

E4 (equal D, different primitive systems). $x_{13} = 4225/64$, $y_{13} = 4225/8$: $D = \{65\}$, $|\mathrm{Gr}| = 2$ versus 4 (Paper III, Theorem 3.6(i)).

E5 (local counts do not determine $\Delta\beta$).

event	$f \to f'$	new cells	$\beta \to \beta'$	$\Delta\beta$	cl
$D = \{385, 27720\}, N = \{11, 72\}$	$(2,1,0) \to (4,3,0)$	$(2,2,0)$	$(1,0,0) \to (1,0,0)$	$(0,0,0)$	$1 \to 1$
$D = \{15, 315\}, N = \{4, 21\}$	$(2,1,0) \to (4,3,0)$	$(2,2,0)$	$(1,0,0) \to (2,1,0)$	$(1,1,0)$	$1 \to 2$
$D = \{12, 35\}, N = \{44, 45\}$	$(2,0,0) \to (4,3,0)$	$(2,3,0)$	$(2,0,0) \to (1,0,0)$	$(-1,0,0)$	$2 \to 1$
$D = \{33, 280\}, N = \{14, 20\}$	$(2,0,0) \to (4,3,0)$	$(2,3,0)$	$(2,0,0) \to (2,1,0)$	$(0,1,0)$	$2 \to 2$

E6 (the scaling signature $(G, \mathbf{period}, \mathbf{multiplier}, \rho)$ does not determine topology). 126 and 3094 (Table 4).

E7 (no one-step correspondence). $C(x)/x \le 2 < 4 \le R(N)$ (Paper III, Proposition 13.6).

F Finite-exact information

These are the numerical results needed for the synthesis comparisons, with the exact checked domain. All computations use exact integer or rational arithmetic (`scripts/`, results in `results/`); none is extrapolated.

Used in	Statement and checked domain	Script
Lemma 8.1	components of $K(D)$ equal share-a-prime classes: 3000 random sets $D \subset [2, 400)$, $\lvert D\rvert \in [2, 9]$; 0 violations	`c5_checks.py`
Proposition 5.3(b)	edge persistence rule $D \subseteq D'$: 28,179 pairs, 0 violations	`c5_checks.py`
Proposition 3.4(b)	the fibre example $c = (7, 0, 0)$, $c' = (0, 2, 4)$ from source 1: equal multiplier 16384, no admissible first letter for c' (witness W10)	`c6_witnesses.py`
Proposition 6.3	$\nu(m)$ formula of Paper II, Theorem 5.8: all sources $1 \le m < 1500$ (1356 with $\nu = 0$, 107 with $\nu = 1$, 36 with $\nu = 2$); 0 violations; 16 histories $1 \to 5184$	`c5_checks.py`, `c6_witnesses.py`
Proposition 7.1	histories/classes for $900, 9000, 288, 4320$ at source 4; census $16 \le E \le 120000$: 238 occurrence sets, 166 shared, 106 with different class counts, 158 with different history counts	`c6_witnesses.py`, `c1_collide.py`
Proposition 8.2(d)	four set-level Betti-change witnesses ($\Delta\beta_0 = \pm 1$, $\Delta\beta_1 = \pm 1$; witness W9); the census of Paper III, Theorem 11.6 is quoted, not recomputed	`c6_witnesses.py`
Proposition 9.1	four events (Appendix E, E5); random search 60,000 events with 2-cells: 2039 keys $(\lambda, \Delta\beta_0)$, no pair with different $(\Delta\beta_1, \Delta\beta_2)$	`c6_witnesses.py`, `c7_dim2.py`
Proposition 10.1	orbit law $C^3(x_k) = 4x_k$, $L = S_0 2^k$, $G = 5$ for $0 \le k \le 11$, $S_0 \in \{126, 3094\}$; Betti numbers $0 \le k \le 14$ (Table 4)	`c4_pair.py`, `c6_witnesses.py`
Proposition 11.3(c)	translation formula and $D(x) \subseteq D(Cx)$: 1008 integer states $x \le 20000$ with integer $C(x)$ (4000 sampled); $N + 1 \in D(T_N(m))$ for all $m < 400$	`c6_witnesses.py`
Proposition 4.1(c)	$D(x_{13}) = D(y_{13}) = \{65\}$ recomputed	`figs_a.py`, `c6_witnesses.py`

k	$\beta(K(D(x_k)))$, $S_0 = 126$	$S_0 = 3094$
0	(1,1)	(1,1)
1	(1,1)	(1,3)
2	(1,4)	(1,10)
3	(1,6)	(1,10)
4	(1,6,1)	(1,10,1)
5	(1,6,1)	(1,10,2)
6	(1,6,3)	(1,10,10)
7	(1,6,5)	(1,10,9)
8	(1,6,5)	(1,10,9)
9	(1,6,5)	(1,10,9)
10	(1,6,5)	(1,10,9,1)
11	(1,6,5)	(1,10,9,1)
12	(1,6,5)	(1,10,9,1)
13	(1,6,5)	(1,10,9,1)
14	(1,6,5)	(1,10,9,1)

Table 4: **Betti numbers of the two $G = 5$ towers** $K(D(x_k))$, $x_k = (S_0 2^k)^2/5$, over $\mathbb{F}_2$, trailing zeros omitted. Finite-exact for $0 \le k \le 14$.

Inherited results. Computations quoted from Papers I–III (the census of Paper III, Theorem 11.6, the Betti numbers of Paper III, Theorem 12.10, and the specimen of Paper I,

Proposition 6.3) carry their original statuses and were not re-run; inherited verification remains NOT_RERUN.